A Conditional Positive-Loss Certificate for Overparameterized Tensor Gradient Descent

July 16, 2026

1 Theoretical Setup

1.1 Smoothed tensor and objective

Fix $\kappa \geq 1$, integers n, r, k, and a fixed $q \geq 4$. Let $\bar{A}, \bar{B}, \bar{C} \in \mathbb{R}^{n\times r}$ be deterministic, with columns $\bar{a}_j, \bar{b}_j, \bar{c}_j$. For a mode matrix $\bar{M}$, write

$$D_M = \operatorname{diag}(\|\bar{m}_1\|_2, \ldots, \|\bar{m}_r\|_2), \qquad \bar{M}^\circ = \bar{M} D_M^{-1}.$$

Independently over modes and columns, draw $\xi_j^a, \xi_j^b, \xi_j^c \sim \mathcal{N}(0, r^{-2q} I_n/n)$ and set

$$a_j = \bar{a}_j + \xi_j^a, \qquad b_j = \bar{b}_j + \xi_j^b, \qquad c_j = \bar{c}_j + \xi_j^c.$$

Let $A = [a_1 \ \cdots \ a_r]$, and define B, C analogously. With

$$D_r = \sum_{j=1}^{r} e_j \otimes e_j \otimes e_j,$$

the realized target is

$$T = \sum_{j=1}^{r} a_j \otimes b_j \otimes c_j = (A \otimes B \otimes C) D_r.$$

For $X = [x_1 \ \cdots \ x_k]$, $Y = [y_1 \ \cdots \ y_k]$, and $Z = [z_1 \ \cdots \ z_k]$, put

$$S(X,Y,Z) = \sum_{i=1}^{k} x_i \otimes y_i \otimes z_i, \qquad F(X,Y,Z) = \|T - S(X,Y,Z)\|_F^2.$$

Under the fixed matricization convention, the Khatri–Rao columns in the X, Y, Z updates are respectively $z_i \otimes y_i$, $z_i \otimes x_i$, and $y_i \otimes x_i$.

For a raw component $(\widetilde{x}, \widetilde{y}, \widetilde{z})$ with three positive norms, let

$$g = (\|\widetilde{x}\|_2 \|\widetilde{y}\|_2 \|\widetilde{z}\|_2)^{1/3}$$

and replace the component by

$$\left(g\frac{\widetilde{x}}{\|\widetilde{x}\|_2}, g\frac{\widetilde{y}}{\|\widetilde{y}\|_2}, g\frac{\widetilde{z}}{\|\widetilde{z}\|_2} \right).$$

If one factor is zero, leave the raw triple unchanged. Denote this columnwise, product-preserving map by $\mathcal{G}$. Draw the entries of $X_0^{\mathrm{raw}}, Y_0^{\mathrm{raw}}, Z_0^{\mathrm{raw}}$ independently from $\mathcal{N}(0, 1/n)$, independently of the smoothing, and set $\theta_0 = (X_0, Y_0, Z_0) = \mathcal{G}(X_0^{\mathrm{raw}}, Y_0^{\mathrm{raw}}, Z_0^{\mathrm{raw}})$. For $\theta_t = (X_t, Y_t, Z_t)$, perform the simultaneous full-batch updates

$$\begin{aligned}
\widetilde{X}_{t+1} &= X_t - \eta \nabla_X F(\theta_t), \\
\widetilde{Y}_{t+1} &= Y_t - \eta \nabla_Y F(\theta_t), \\
\widetilde{Z}_{t+1} &= Z_t - \eta \nabla_Z F(\theta_t), \\
\theta_{t+1} &= \mathcal{G}(\widetilde{X}_{t+1}, \widetilde{Y}_{t+1}, \widetilde{Z}_{t+1}), \qquad \eta = (nkr)^{-12}.
\end{aligned}$$

There is no projection, clipping, weight decay, sparsification, restart, or early stopping.

1.2 Coefficient geometry and generated events

On the event that A, B, C have full column rank, define

$$\alpha_{i,t} = A^\dagger x_{i,t}, \qquad \beta_{i,t} = B^\dagger y_{i,t}, \qquad \gamma_{i,t} = C^\dagger z_{i,t},$$

and normalized initial coordinates

$$\bar{\alpha}_{i,0} = \sqrt{n/r}\,\alpha_{i,0}, \quad \bar{\beta}_{i,0} = \sqrt{n/r}\,\beta_{i,0}, \quad \bar{\gamma}_{i,0} = \sqrt{n/r}\,\gamma_{i,0}.$$

Set

$$\widehat{D}_0 = \sum_{i=1}^{k} \alpha_{i,0} \otimes \beta_{i,0} \otimes \gamma_{i,0}, \qquad \delta_0 = \frac{1}{8}.$$

The raw tangent span is

$$\begin{aligned}
\mathscr{S}_0^{\mathrm{raw}} = \operatorname{span}\{&u \otimes \beta_{i,0} \otimes \gamma_{i,0}, \alpha_{i,0} \otimes v \otimes \gamma_{i,0}, \alpha_{i,0} \otimes \beta_{i,0} \otimes w : \\
&u, v, w \in \mathbb{R}^r,\ i \in [k]\},
\end{aligned}$$

and $\mathscr{S}_0^{\mathrm{norm}}$ is defined by replacing fixed coefficient vectors by their barred versions. Multiplication of each generating block by its nonzero normalization scalar leaves its range unchanged, so $\mathscr{S}_0^{\mathrm{raw}} = \mathscr{S}_0^{\mathrm{norm}}$; this definition-level identity is verified explicitly in Proposition A.2. Write their common value as $\mathscr{S}_0$. Normalization is used only for the certificate geometry: D_r, $\widehat{D}_0$, and the physical residual remain raw.

Let $\kappa_1 = 2\kappa^2$. Define the realized-factor event

$$\mathcal{E}_{\mathrm{cond}} = \bigcap_{M \in \{A,B,C\}} \{\|M\|_{\mathrm{op}} \le \kappa_1,\ \sigma_{\min}(M) \ge \kappa_1^{-1}\}.$$

For the normalized pair matrices, for example

$$\bar{K}_0^{\bar{\beta}\bar{\gamma}} = [\bar{\beta}_{1,0} \otimes \bar{\gamma}_{1,0} \ \cdots \ \bar{\beta}_{k,0} \otimes \bar{\gamma}_{k,0}],$$

let $\mathcal{E}_{\mathrm{gram}}^{\mathrm{norm}}$ be the event that every eigenvalue of each of the three normalized pair Grams lies in $[r^{-20}, r^{20}]$. The raw and normalized pair Grams obey the exact identity

$$G_{\mathrm{raw}}^{pq} = (r/n)^2 G_{\mathrm{norm}}^{pq} \quad (pq \in \{\beta\gamma, \alpha\gamma, \alpha\beta\}).$$

Define

$$\begin{aligned}
\mathcal{E}_{\mathrm{deficit}} = \{\exists W_0 : \|W_0\|_F = 1,\ &W_0 \perp \mathscr{S}_0, \\
&\langle D_r - \widehat{D}_0, W_0 \rangle_F \ge \delta_0 \|D_r\|_F\},
\end{aligned}$$

and

$$\mathcal{E}_{\mathrm{size}} = \left\{ \max_{i\in[k],\, m\in\{x,y,z\}} \|m_{i,0}\|_2 \le 2 \right\}.$$

The generated initialization event is

$$\mathcal{E}_{\mathrm{init\_norm}} = \mathcal{E}_{\mathrm{cond}} \cap \mathcal{E}_{\mathrm{gram}}^{\mathrm{norm}} \cap \mathcal{E}_{\mathrm{deficit}} \cap \mathcal{E}_{\mathrm{size}}.$$

Define the raw coefficient map

$$\Psi_{A,B,C}(X,Y,Z) = \sum_{i=1}^{k} (A^\dagger x_i) \otimes (B^\dagger y_i) \otimes (C^\dagger z_i),$$

and the balanced-representative distance and total path length

$$\begin{aligned} d_{\mathrm{bal}}(\theta,\theta') &= \left(\|X-X'\|_F^2 + \|Y-Y'\|_F^2 + \|Z-Z'\|_F^2\right)^{1/2}, \\ E_{\mathrm{path}} &= \sum_{t\ge 0} d_{\mathrm{bal}}(\theta_{t+1},\theta_t). \end{aligned}$$

With

$$C_{\mathrm{CP}}(\kappa,R) = \kappa_1^3(1+3R), \qquad E_\star = \min\left\{1, \sqrt{\frac{\delta_0}{16 C_{\mathrm{CP}}(\kappa,3)}}\right\},$$

the sole trajectory certificate is

$$\mathcal{C}_{\mathrm{path}} = \{E_{\mathrm{path}} \le E_\star\}.$$

It contains no convergence, boundedness, Gram, trapping, or positive-loss clause. Finally, with $P_M = MM^\dagger$,

$$(P_A \otimes P_B \otimes P_C)(T - S(\theta)) = (A \otimes B \otimes C)(D_r - \Psi_{A,B,C}(\theta)).$$

1.3 Assumptions

Assumption 1 (Well-conditioned deterministic bases)**.** *Every base column is nonzero and satisfies* $\kappa^{-1} \le \|\bar m_j\|_2 \le \kappa$*. For* $M \in \{\bar A, \bar B, \bar C\}$*, every singular value of* M° *lies in* $[\kappa^{-1},\kappa]$*.*

Assumption 2 (Smoothed dimension regime)**.** *The integer* $q \ge 4$ *is fixed,* r *is sufficiently large for the asymptotic statement, and* $n \ge C(\kappa,q) r^4 \log r$*, with no upper restriction on* n*.*

Assumption 3 (Universal superlinear rank window)**.** *The exponent is* $c = 1/4$ *and* $r < k \le \lfloor r^{5/4} \rfloor$*.*

Assumption 4 (Independent Gaussian smoothing)**.** *All smoothing vectors are mutually independent with law* $\mathcal{N}(0, r^{-2q} I_n/n)$ *and are independent of initialization.*

Assumption 5 (Gaussian initialization)**.** *Before balancing, all entries of* $X_0^{\mathrm{raw}}, Y_0^{\mathrm{raw}}, Z_0^{\mathrm{raw}}$ *are independent* $\mathcal{N}(0,1/n)$ *variables.*

Assumption 6 (Fixed balanced gradient-descent protocol)**.** *The simultaneous full-batch update, step size* $\eta = (nkr)^{-12}$*, and product-preserving map* $\mathcal{G}$ *are exactly those specified above, with no additional optimization operation.*

1.4 Formalized goal

The objective is a conditional theorem. Uniformly over the stated dimension and rank ranges and every admissible deterministic base triple, the generated event must satisfy

$$\mathbb{P}(\mathcal{E}_{\text{init\_norm}}) \geq 1 - r^{-10}.$$

On $\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}$, the balanced iterates must converge to a finite θ_∞ and have a strictly positive relative limiting loss. If $\mathcal{F}_+$ denotes this convergence-and-positive-limit event, the required probability statement is

$$\begin{aligned}\mathbb{P}(\mathcal{F}_+) &\geq \mathbb{P}(\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}) \\ &= \mathbb{P}(\mathcal{E}_{\text{init\_norm}})\mathbb{P}(\mathcal{C}_{\text{path}} \mid \mathcal{E}_{\text{init\_norm}}) \\ &\geq (1 - r^{-10})\mathbb{P}(\mathcal{C}_{\text{path}} \mid \mathcal{E}_{\text{init\_norm}}).\end{aligned}$$

No positive lower bound for the final conditional factor is claimed. Proving such a uniform lower bound remains the source-level gap; in particular, the result below does not solve the original unconditional open problem.

2 Preliminaries

All tensor spaces carry their Euclidean, equivalently Frobenius, inner products. Orthogonal projections are denoted by P_E, and tensor products of mode maps act in the order fixed in Section 1. The notation $\kappa_1 = 2\kappa^2$, $C_{\text{CP}}(\kappa, R)$, $E_\star$, and the events $\mathcal{E}_{\text{init\_norm}}$ and $\mathcal{C}_{\text{path}}$ are those defined in the setup. They are the only auxiliary quantities needed in the theorem statement. All orientation, quotient, truncation, and Taylor-expansion objects used in the proof are introduced locally in the appendix.

3 Main Theorem

Theorem 3.1 (Conditional positive-loss certificate)**.** *Under Assumptions 1–6, there exist $r_0(\kappa, q)$ and $C(\kappa, q)$ such that, uniformly for every*

$$r \geq r_0(\kappa, q), \qquad n \geq C(\kappa, q) r^4 \log r, \qquad r < k \leq \lfloor r^{5/4} \rfloor,$$

and every deterministic base triple satisfying Assumption 1,

$$\mathbb{P}(\mathcal{E}_{\text{init\_norm}}) \geq 1 - r^{-10}.$$

Define

$$\epsilon_0(\kappa) = \left(\frac{15}{16}\delta_0\right)^2 \kappa_1^{-12} > 0.$$

On the explicitly conditional event $\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}$, the balanced iterates converge in d_{bal} to a finite θ_∞ and satisfy

$$\lim_{t \to \infty} F(\theta_t) = F(\theta_\infty) \geq \epsilon_0(\kappa) \|T\|_F^2 > 0.$$

Consequently, for the event $\mathcal{F}_+$ defined in Section 1,

$$\mathbb{P}(\mathcal{F}_+) \geq (1 - r^{-10})\mathbb{P}(\mathcal{C}_{\text{path}} \mid \mathcal{E}_{\text{init\_norm}}).$$

The last conditional factor is retained exactly and may be zero; no uniform positive lower bound for it, and hence no unconditional positive-probability failure theorem, is asserted.

The exposed quantities are n, r, k, κ, q, the confidence $1-r^{-10}$, the explicit relative-loss constant $\epsilon_0(\kappa)$, and the displayed conditional path probability. The constants r_0, C may depend only on κ, q. Probability is under the joint smoothing and initialization law conditional on the deterministic base triple; the horizon is all time with an asymptotic endpoint; and the norms are precisely d_{bal} and the physical tensor Frobenius norm. At zero path length, the endpoint equals the initialization and the proof retains the stronger full initial margin.

4 Proof Sketch

The proof first establishes realized conditioning and rewrites the balanced coefficient initialization as elliptic images of independent standard Gaussian arrays. Radius control and a rectangular Khatri–Rao concentration argument yield the normalized pair-Gram event, while direct algebra preserves the raw tangent span and the raw represented tensor.

Positive-diagonal QR separates each Gaussian mode array into an orientation and a complete internal shape. After absorbing each determinant bit into the shape, the three orientations are independent Haar matrices in $SO(r)$ and are independent of a measurable internal subspace of dimension at most $3kr$. Sequential Haar twirling gives projection mean d/r^3 for a fixed target. An explicit product-metric Lipschitz estimate and a metric-normalized Ricci–log-Sobolev–Herbst calculation give the required product-Haar tail. The exact oblique-basis projector transfers this estimate through the elliptic maps with loss κ_1^{12}. At the maximal rank in Assumption 3, this yields

$$\|P_{\mathscr{S}_0} D_r\|_F^2 \le r/2.$$

The normalized orthogonal component of the raw D_r is therefore a unit witness for $\mathcal{E}_{\mathrm{deficit}}$. Together with the conditioning, Gram, and size estimates, a conditional union bound proves the initialization confidence.

On the sole trajectory certificate $\mathcal{C}_{\mathrm{path}}$, finite total variation makes the balanced representatives Cauchy and bounds every factor column by 3. A direct trilinear expansion gives a single endpoint remainder at most $\delta_0/16$. Orthogonality of the witness cancels the first-order term and leaves raw coefficient margin at least $(15/16)\delta_0\sqrt{r}$. The exact same-target projection identity then gives the physical relative floor

$$F(\theta_\infty) \ge \left(\frac{15}{16}\delta_0\right)^2 \kappa_1^{-12}\|T\|_F^2.$$

Polynomial continuity of the CP objective and elementary event conditioning complete Theorem 3.1, while retaining the possibly zero factor $\mathbb{P}(\mathcal{C}_{\mathrm{path}} \mid \mathcal{E}_{\mathrm{init\_norm}})$.

A Proof Details

A.1 Realized-factor conditioning

Lemma A.1 (conditioning of the deterministic base matrices)**.** *Under Assumption 1, for every $\bar{M} \in \{\bar{A}, \bar{B}, \bar{C}\}$,*

$$\sigma_{\min}(\bar{M}) \geq \kappa^{-2}, \qquad \|\bar{M}\|_{\mathrm{op}} \leq \kappa^2.$$

Proof. Write, exactly as in the setting,

$$\bar{M} = \bar{M}^{\circ} D_M, \qquad D_M = \mathrm{diag}(\|\bar{m}_1\|_2, \dots, \|\bar{m}_r\|_2).$$

Assumption 1 makes D_M invertible and gives

$$\sigma_{\min}(\bar{M}^{\circ}) \geq \kappa^{-1}, \quad \|\bar{M}^{\circ}\|_{\mathrm{op}} \leq \kappa, \quad \sigma_{\min}(D_M) \geq \kappa^{-1}, \quad \|D_M\|_{\mathrm{op}} \leq \kappa.$$

For every $x \in \mathbb{R}^r$,

$$\|\bar{M}x\|_2 = \|\bar{M}^{\circ} D_M x\|_2 \geq \sigma_{\min}(\bar{M}^{\circ}) \|D_M x\|_2 \geq \kappa^{-2} \|x\|_2,$$

which proves the lower singular bound. Submultiplicativity gives

$$\|\bar{M}\|_{\mathrm{op}} \leq \|\bar{M}^{\circ}\|_{\mathrm{op}} \|D_M\|_{\mathrm{op}} \leq \kappa^2.$$

This proves the lemma. □

Lemma A.2 (simultaneous Gaussian perturbation control)**.** *Under Assumptions 2 and 4, define*

$$r_{0,\mathrm{cond}}(\kappa, q) := \max\left\{3, \left\lceil (6\kappa^2)^{1/q} \right\rceil \right\}.$$

Choose the dimension constant in Assumption 2 so that $C(\kappa, q) \geq 1$. Then, for every integer $r \geq r_{0,\mathrm{cond}}(\kappa, q)$ and every $n \geq C(\kappa, q) r^4 \log r$, the event

$$\mathcal{E}_{\mathrm{pert}} := \left\{ \max_{M \in \{A,B,C\}} \|M - \bar{M}\|_{\mathrm{op}} \leq \frac{1}{2\kappa^2} \right\}$$

satisfies

$$\mathbb{P}(\mathcal{E}_{\mathrm{pert}}) \geq 1 - r^{-20}.$$

The probability is uniform over all deterministic base triples satisfying the setting assumptions and over all such n.

Proof. Put $\rho = r^{-q}$, $\Xi_M = M - \bar{M}$, and

$$u_r = \sqrt{2 \log(3r^{20})}.$$

The rectangular Gaussian norm bound of Davidson and Szarek (2001, Theorem II.13) states, in the present notation, that an $n \times r$ matrix G with iid $\mathcal{N}(0,1)$ entries satisfies

$$\mathbb{P}\big(\|G\|_{\mathrm{op}} > \sqrt{n} + \sqrt{r} + u\big) \leq e^{-u^2/2}, \qquad u \geq 0.$$

Assumption 4 gives $\Xi_M = (\rho/\sqrt{n}) G_M$ with such a G_M. Hence, for each of the three modes,

$$\mathbb{P}\Big(\|\Xi_M\|_{\mathrm{op}} > \rho\left(1 + \sqrt{r/n} + u_r/\sqrt{n}\right)\Big) \leq \frac{1}{3r^{20}}. \tag{A.1}$$

For $r \geq 3$,

$$u_r^2 = 2\log 3 + 40\log r \leq 42\log r \leq r^4 \log r \leq n, \tag{A.2}$$

where the penultimate inequality uses $r^4 \geq 81 > 42$. Also $n \geq r^4 \log r \geq r$. Hence

$$\sqrt{r/n} \leq 1, \qquad u_r/\sqrt{n} \leq 1. \tag{A.3}$$

Combining (A.1)–(A.3), outside a per-mode event of probability at most $(3r^{20})^{-1}$,

$$\|\Xi_M\|_{\mathrm{op}} \leq 3\rho = 3r^{-q}.$$

The definition of $r_{0,\mathrm{cond}}(\kappa, q)$ gives

$$3r^{-q} \leq \frac{1}{2\kappa^2}. \tag{A.4}$$

A union bound over $M = A, B, C$, which does not require independence, yields

$$\mathbb{P}(\mathcal{E}_{\mathrm{pert}}^c) \leq 3 \cdot \frac{1}{3r^{20}} = r^{-20}. \tag{A.5}$$

The right sides of (A.1)–(A.5) do not depend on the deterministic base triple. Increasing n only decreases $\sqrt{r/n}$ and $u_r/\sqrt{n}$, so the argument has no upper-dimensional restriction. This proves the lemma.

□

Proposition A.1 (realized-factor conditioning)**.** *Under Assumptions 1, 2, and 4, let $\kappa_1 = 2\kappa^2$, choose $C(\kappa, q) \geq 1$, and suppose $r \geq r_{0,\mathrm{cond}}(\kappa, q)$. Then, simultaneously for $M \in \{A, B, C\}$, with probability at least $1 - r^{-20}$,*

$$\sigma_{\min}(M) \geq \kappa_1^{-1}, \qquad \|M\|_{\mathrm{op}} \leq \kappa_1, \qquad \|M^\dagger\|_{\mathrm{op}} \leq \kappa_1.$$

In particular, the generated event $\mathcal{E}_{\mathrm{cond}}$ holds with probability at least $1 - r^{-20}$.

Proof. For same-size matrices X, E, Weyl's singular-value perturbation inequality in the form

$$\sigma_{\min}(X + E) \geq \sigma_{\min}(X) - \|E\|_{\mathrm{op}}$$

is the standard current-notation specialization of Stewart and Sun (1990, Chapter IV). Apply it with $X = \bar{M}$, $E = M - \bar{M}$. On $\mathcal{E}_{\mathrm{pert}}$, together with Lemma A.1, this gives, for every mode,

$$\sigma_{\min}(M) \geq \kappa^{-2} - \frac{1}{2\kappa^2} = \frac{1}{2\kappa^2} = \kappa_1^{-1}. \tag{A.6}$$

The operator-norm triangle inequality gives

$$\|M\|_{\mathrm{op}} \leq \kappa^2 + \frac{1}{2\kappa^2} \leq 2\kappa^2 = \kappa_1, \tag{A.7}$$

where the second inequality uses $\kappa \geq 1$. Equation (A.6) makes every realized mode matrix full column rank and therefore

$$\|M^\dagger\|_{\mathrm{op}} = \sigma_{\min}(M)^{-1} \leq \kappa_1. \tag{A.8}$$

Equations (A.6)–(A.7) show that $\mathcal{E}_{\mathrm{pert}} \subseteq \mathcal{E}_{\mathrm{cond}}$. By Lemma A.2, $\mathbb{P}(\mathcal{E}_{\mathrm{pert}}) \geq 1 - r^{-20}$, so the same lower bound holds for $\mathbb{P}(\mathcal{E}_{\mathrm{cond}})$. This proves the proposition.

□

Proof of the subsection conclusion. Lemma A.1 converts Assumption 1 into the deterministic interval $[\kappa^{-2}, \kappa^2]$ for each base mode matrix. Lemma A.2 uses the exact smoothing variance r^{-2q}/n, the explicit threshold $u_r = \sqrt{2\log(3r^{20})}$, and a union over the three modes to produce $\mathcal{E}_{\mathrm{pert}}$ with failure at most r^{-20}, uniformly over all admissible bases and all n above the lower dimension threshold. Proposition A.1 then applies Weyl to obtain, with $\kappa_1 = 2\kappa^2$,

$$\sigma_{\min}(M) \geq \kappa_1^{-1}, \qquad \|M\|_{\mathrm{op}} \leq \kappa_1 \quad (M = A, B, C),$$

on an event of probability at least $1 - r^{-20}$. These are precisely the conditions defining $\mathcal{E}_{\mathrm{cond}}$, which proves the proposition. No initialization-geometry, tangent, or trajectory claim is used or proved here.

□

A.2 Balanced coefficient Gaussianization

For the raw initialization matrices from Section 1, write

$$X_0^{\mathrm{raw}} = [\widetilde{x}_1 \ \cdots \ \widetilde{x}_k], \qquad Y_0^{\mathrm{raw}} = [\widetilde{y}_1 \ \cdots \ \widetilde{y}_k], \qquad Z_0^{\mathrm{raw}} = [\widetilde{z}_1 \ \cdots \ \widetilde{z}_k],$$

and pair the factor matrices, physical modes, and coefficient coordinates as

$$(A, x, \alpha), \qquad (B, y, \beta), \qquad (C, z, \gamma).$$

Thus $m = x, y, z$ when $M = A, B, C$, respectively. Define the pre-balancing coefficient vectors by

$$\widetilde{\zeta}_i^A = A^\dagger \widetilde{x}_i, \qquad \widetilde{\zeta}_i^B = B^\dagger \widetilde{y}_i, \qquad \widetilde{\zeta}_i^C = C^\dagger \widetilde{z}_i.$$

After balancing, write

$$\zeta_i^A = \alpha_{i,0}, \qquad \zeta_i^B = \beta_{i,0}, \qquad \zeta_i^C = \gamma_{i,0}, \qquad \bar{\zeta}_i^M = \sqrt{\frac{n}{r}}\,\zeta_i^M, \quad M \in \{A, B, C\}.$$

Lemma A.3 (Gaussian radii and exact balancing scalars)**.** *Under Assumption 5, define, for every component i,*

$$g_i^x = \sqrt{n}\,\widetilde{x}_i, \qquad g_i^y = \sqrt{n}\,\widetilde{y}_i, \qquad g_i^z = \sqrt{n}\,\widetilde{z}_i, \qquad \chi_i^m = \|g_i^m\|_2.$$

Then the g_i^m are independent $\mathcal{N}(0, I_n)$ vectors over all i and m. Almost surely, every χ_i^m is positive, the directions $\omega_i^m = g_i^m/\chi_i^m$ are independent uniform sphere vectors and are independent of all radii, and the balancing map satisfies

$$x_{i,0} = s_i^x \widetilde{x}_i, \qquad y_{i,0} = s_i^y \widetilde{y}_i, \qquad z_{i,0} = s_i^z \widetilde{z}_i,$$

where, on the positive-radius branch,

$$s_i^x = \left(\frac{\chi_i^y \chi_i^z}{(\chi_i^x)^2}\right)^{1/3}, \quad s_i^y = \left(\frac{\chi_i^x \chi_i^z}{(\chi_i^y)^2}\right)^{1/3}, \quad s_i^z = \left(\frac{\chi_i^x \chi_i^y}{(\chi_i^z)^2}\right)^{1/3}.$$

On the setting's zero-factor branch, define $s_i^x = s_i^y = s_i^z = 1$. With this extension every scalar is nonzero, the displayed scalar representation agrees with $\mathcal{G}$ on every branch, and

$$s_i^x s_i^y s_i^z = 1$$

identically.

Proof. Assumption 5 gives the stated joint standard Gaussian law after multiplication by $\sqrt{n}$. The density of one g_i^m is a function only of its Euclidean radius. In polar coordinates its measure is proportional to

$$e^{-\chi^2/2}\chi^{n-1}\,d\chi\,d\sigma(\omega),$$

where $d\sigma$ is uniform surface measure. This factorization proves the radius-direction independence and uniformity. The singleton $\{0\}$ has Gaussian measure zero; a finite union over the $3k$ raw columns therefore shows that all radii are positive almost surely.

On that probability-one branch, the three raw physical norms are $\chi_i^x/\sqrt{n}, \chi_i^y/\sqrt{n}, \chi_i^z/\sqrt{n}$, so their geometric mean is

$$q_i = \frac{(\chi_i^x\chi_i^y\chi_i^z)^{1/3}}{\sqrt{n}}.$$

The balancing definition gives $s_i^x = q_i/(\chi_i^x/\sqrt{n})$, and similarly in the other modes, which is exactly the displayed formula. In particular, balancing changes only radii: $x_{i,0} = q_i\omega_i^x$, and analogously for y, z, so all raw directions are preserved. Direct multiplication gives

$$(s_i^x s_i^y s_i^z)^3 = \frac{(\chi_i^y\chi_i^z)(\chi_i^x\chi_i^z)(\chi_i^x\chi_i^y)}{(\chi_i^x\chi_i^y\chi_i^z)^2} = 1.$$

All three scalars are positive, hence their product is one rather than another cube root of one. If any raw factor is zero, the setting leaves the entire raw triple unchanged; assigning all three scalars the value one represents that branch exactly and preserves the product identity. This proves the lemma.

□

Lemma A.4 (compact-SVD Gaussianization of the balanced coefficients)**.** *Under Assumption 5, Proposition A.1, and Lemma A.3, condition on any realized triple* $(A, B, C) \in \mathcal{E}_{\mathrm{cond}}$*. For each* $M \in \{A, B, C\}$*, choose a compact SVD*

$$M = U_M\Sigma_M V_M^\top, \qquad U_M^\top U_M = V_M^\top V_M = I_r, \qquad \Sigma_M \succ 0,$$

and define

$$H_M = V_M\Sigma_M^{-1}, \qquad z_i^M = U_M^\top g_i^m,$$

using the mode pairing $(A, x), (B, y), (C, z)$*. Then, under the conditional initialization law, the full family* $\{z_i^M : M \in \{A, B, C\}, i \in [k]\}$ *is independent with* $z_i^M \sim \mathcal{N}(0, I_r)$*, every* H_M *is invertible and satisfies*

$$\sigma(H_M) \subset [\kappa_1^{-1}, \kappa_1],$$

and the raw, balanced, and normalized coefficient vectors obey the exact identities

$$\widetilde{\zeta}_i^M = \frac{1}{\sqrt{n}}H_M z_i^M, \qquad \zeta_i^M = \frac{s_i^m}{\sqrt{n}}H_M z_i^M, \qquad \boxed{\bar{\zeta}_i^M = \frac{s_i^m}{\sqrt{r}}H_M z_i^M}.$$

No independence between s_i^m *and* z_i^M *is asserted or needed.*

Proof. Conditional on the realized factors, all SVD matrices are deterministic, while the standardized raw initialization vectors remain independent standard Gaussians because initialization is independent of smoothing. Since $U_M^\top U_M = I_r$,

$$\mathbb{E}[z_i^M(z_i^M)^\top \mid A, B, C] = U_M^\top U_M = I_r,$$

and each z_i^M is Gaussian. Distinct (M,i) use distinct independent raw vectors, proving independence across all modes and components.

The compact-SVD pseudoinverse formula gives

$$M^\dagger = V_M \Sigma_M^{-1} U_M^\top.$$

Because $\widetilde{m}_i = g_i^m/\sqrt{n}$,

$$\widetilde{\zeta}_i^M = M^\dagger \widetilde{m}_i = \frac{1}{\sqrt{n}} V_M \Sigma_M^{-1} U_M^\top g_i^m = \frac{1}{\sqrt{n}} H_M z_i^M. \tag{A.9}$$

Lemma A.3 gives $m_{i,0} = s_i^m \widetilde{m}_i$ on every branch under its scalar extension. Linearity of $M^\dagger$ therefore yields

$$\zeta_i^M = s_i^m \widetilde{\zeta}_i^M = \frac{s_i^m}{\sqrt{n}} H_M z_i^M. \tag{A.10}$$

Multiplication by the setting's coefficient normalization $\sqrt{n/r}$ gives the boxed formula, with no omitted factor.

The singular values of $H_M = V_M \Sigma_M^{-1}$ are exactly the reciprocals of the singular values of M. On $\mathcal{E}_{\mathrm{cond}}$, $\sigma(M) \subset [\kappa_1^{-1}, \kappa_1]$, and this interval is invariant under reciprocal, proving the claimed bounds for H_M. Repeated singular values, permutations, or SVD sign choices only apply orthogonal changes to the standard Gaussian coordinates and do not alter their joint law or the singular values of H_M. Thus any fixed compact-SVD convention, including any measurable convention used below, gives the same identities. Finally, the scalars are functions of the full raw radii and may be dependent on the projected Gaussian coordinates; none of the algebraic conclusions requires independence. This proves the lemma.

□

Proposition A.2 (exact tangent-block and coefficient-product invariance)**.** *Under Assumption 5, Proposition A.1, and Lemmas A.3 and A.4, condition on any realized triple in $\mathcal{E}_{\mathrm{cond}}$. For each component define the three elliptic-Gaussian baseline tangent blocks*

$$\begin{aligned}
\mathcal{B}_i^x &= \{u \otimes H_B z_i^B \otimes H_C z_i^C : u \in \mathbb{R}^r\},\\
\mathcal{B}_i^y &= \{H_A z_i^A \otimes v \otimes H_C z_i^C : v \in \mathbb{R}^r\},\\
\mathcal{B}_i^z &= \{H_A z_i^A \otimes H_B z_i^B \otimes w : w \in \mathbb{R}^r\}.
\end{aligned}$$

Then every tangent block in the setting, both raw and normalized, has exactly the corresponding baseline range. More precisely,

$$\begin{aligned}
\{u \otimes \beta_{i,0} \otimes \gamma_{i,0} : u \in \mathbb{R}^r\} &= \frac{s_i^y s_i^z}{n} \mathcal{B}_i^x = \mathcal{B}_i^x,\\
\{\alpha_{i,0} \otimes v \otimes \gamma_{i,0} : v \in \mathbb{R}^r\} &= \frac{s_i^x s_i^z}{n} \mathcal{B}_i^y = \mathcal{B}_i^y,\\
\{\alpha_{i,0} \otimes \beta_{i,0} \otimes w : w \in \mathbb{R}^r\} &= \frac{s_i^x s_i^y}{n} \mathcal{B}_i^z = \mathcal{B}_i^z,
\end{aligned} \tag{A.11}$$

and

$$\begin{aligned}
\{u \otimes \bar{\beta}_{i,0} \otimes \bar{\gamma}_{i,0} : u \in \mathbb{R}^r\} &= \frac{s_i^y s_i^z}{r} \mathcal{B}_i^x = \mathcal{B}_i^x,\\
\{\bar{\alpha}_{i,0} \otimes v \otimes \bar{\gamma}_{i,0} : v \in \mathbb{R}^r\} &= \frac{s_i^x s_i^z}{r} \mathcal{B}_i^y = \mathcal{B}_i^y,\\
\{\bar{\alpha}_{i,0} \otimes \bar{\beta}_{i,0} \otimes w : w \in \mathbb{R}^r\} &= \frac{s_i^x s_i^y}{r} \mathcal{B}_i^z = \mathcal{B}_i^z.
\end{aligned} \tag{A.12}$$

Consequently,

$$\mathscr{S}_0^{\mathrm{raw}} = \mathscr{S}_0^{\mathrm{norm}} = \mathrm{span}_{i\in[k]}(\mathcal{B}_i^x \cup \mathcal{B}_i^y \cup \mathcal{B}_i^z) = \mathscr{S}_0. \tag{A.13}$$

Moreover balancing preserves the exact raw coefficient tensor componentwise:

$$\widehat{D}_0 = \sum_{i=1}^k \alpha_{i,0}\otimes\beta_{i,0}\otimes\gamma_{i,0} = \sum_{i=1}^k \widetilde{\zeta}_i^A \otimes \widetilde{\zeta}_i^B \otimes \widetilde{\zeta}_i^C = \frac{1}{n^{3/2}}\sum_{i=1}^k H_A z_i^A \otimes H_B z_i^B \otimes H_C z_i^C. \tag{A.14}$$

The normalization is certificate-only:

$$\sum_{i=1}^k \bar{\alpha}_{i,0}\otimes\bar{\beta}_{i,0}\otimes\bar{\gamma}_{i,0} = \left(\frac{n}{r}\right)^{3/2}\widehat{D}_0, \tag{A.15}$$

so neither $\widehat{D}_0$ nor the fixed raw target D_r is replaced by a normalized tensor.

Proof. Lemma A.4 gives

$$\beta_{i,0} = \frac{s_i^y}{\sqrt{n}} H_B z_i^B, \qquad \gamma_{i,0} = \frac{s_i^z}{\sqrt{n}} H_C z_i^C.$$

Substitution into the first raw tangent family yields the first equality in (A.11). The other two follow by the same substitution in their displayed modes. The normalized coefficient formula replaces each denominator $\sqrt{n}$ by $\sqrt{r}$, giving (A.12). By Lemma A.3, every pair product appearing in (A.11)–(A.12) is nonzero; also $n, r > 0$. Multiplication of a linear subspace by a nonzero scalar is a bijection of that subspace, proving the second equality in each line. Taking the span over all components and all three mode blocks proves (A.13). This argument remains valid when a displayed baseline block happens to be the zero subspace.

For each component, the same Gaussianization and the exact product-one identity give

$$\begin{aligned}
\alpha_{i,0}\otimes\beta_{i,0}\otimes\gamma_{i,0} &= \frac{s_i^x s_i^y s_i^z}{n^{3/2}} H_A z_i^A \otimes H_B z_i^B \otimes H_C z_i^C \\
&= \frac{1}{n^{3/2}} H_A z_i^A \otimes H_B z_i^B \otimes H_C z_i^C \\
&= \widetilde{\zeta}_i^A \otimes \widetilde{\zeta}_i^B \otimes \widetilde{\zeta}_i^C.
\end{aligned}$$

Summing proves (A.14). Repeating the calculation with the normalized formula gives the factor $r^{-3/2}$, which compared with (A.14) is exactly $(n/r)^{3/2}$, proving (A.15). The target D_r is fixed by the realized factor model and is not an initialization output, so neither balancing nor the coefficient normalization acts on it. Thus the tangent range and the raw coefficient target remain exactly the objects specified in Section 1.

□

Proof of the subsection conclusion. Lemma A.3 derives the exact raw radius-direction law and the balancing multipliers from Assumption 5. It proves that the zero branch is null, supplies an exact scalar extension on that branch, and proves that all balancing scalars are nonzero with $s_i^x s_i^y s_i^z = 1$.

Conditional on $\mathcal{E}_{\mathrm{cond}}$, Lemma A.4 applies compact SVDs to the raw initialization and proves, with all factors displayed,

$$\bar{\zeta}_i^M = \frac{s_i^m}{\sqrt{r}} H_M z_i^M, \qquad z_i^M \overset{\mathrm{iid}}{\sim} \mathcal{N}(0, I_r), \qquad \sigma(H_M) \subset [\kappa_1^{-1}, \kappa_1].$$

The independence is simultaneous across every mode and component, conditional on the realized factors. The lemma explicitly records that the balancing scalars may depend on the Gaussian arrays, which creates no gap because the remaining conclusions are exact scalar identities.

Proposition A.2 then shows that every individual tangent block is multiplied only by a nonzero pair scalar, including the fixed $1/n$ or $1/r$ normalization factor. Hence every block range, and therefore $\mathscr{S}_0$, is unchanged. The same proposition uses the triple-product identity to prove exact componentwise and summed preservation of $\widehat{D}_0$, while keeping D_r and the raw residual convention untouched. Together these results give the exact H_M, z_i^M, s_i^m representation and its invariance properties. No Gram concentration, quotient-range statement, Haar factorization, or Haar disintegration is asserted here.

□

A.3 Normalized Khatri–Rao Gram control

Lemma A.5 (full-radius control and dependent balancing diagonals)**.** *Under Assumptions 2, 3, and 5, enlarge the dimension constant so that $C(\kappa, q) \geq 1$, and suppose $r \geq 2^{40}$. Define the proof-local event*

$$\mathcal{R}_{\mathrm{rad}} := \bigcap_{i=1}^{k} \bigcap_{m \in \{x,y,z\}} \left\{ \frac{1}{2} \leq \frac{(\chi_i^m)^2}{n} \leq \frac{3}{2} \right\}.$$

Then, conditional on any realized smoothing outcome,

$$\mathbb{P}(\mathcal{R}_{\mathrm{rad}}^c \mid A, B, C) \leq r^{-28}.$$

On $\mathcal{R}_{\mathrm{rad}}$, for every component i,

$$3^{-1/3} \leq s_i^y s_i^z \leq 3^{1/3}, \qquad 3^{-1/3} \leq s_i^x s_i^z \leq 3^{1/3}, \qquad 3^{-1/3} \leq s_i^x s_i^y \leq 3^{1/3}.$$

Proof. Initialization is independent of smoothing, so, after conditioning on (A, B, C), the $3k$ vectors g_i^m remain independent $\mathcal{N}(0, I_n)$. The Laurent–Massart inequalities (Laurent and Massart, 2000, Lemma 1) imply, for $g \sim \mathcal{N}(0, I_d)$ and $0 < \delta < 1$,

$$\mathbb{P}\left(\left|\|g\|_2^2/d - 1\right| > \delta\right) \leq 2\exp(-d\delta^2/8).$$

Indeed, substitute $x = d\delta^2/8$ into their upper and lower chi-square deviations; for $0 < \delta < 1$, both resulting thresholds lie inside the deviation $d\delta$. Applying this with $d = n$, $\delta = 1/2$, and a union bound gives

$$\mathbb{P}(\mathcal{R}_{\mathrm{rad}}^c \mid A, B, C) \leq 6ke^{-n/32} \leq 6r^{5/4}e^{-r^4 \log r/32}.$$

For $r \geq 2^{40}$, one has $6 \leq r^{3/4}$ and $r^4/32 \geq 30$, hence

$$6r^{5/4}e^{-r^4 \log r/32} \leq r^2 r^{-30} = r^{-28}. \tag{A.16}$$

This bound improves as n increases, so it is uniform with no upper restriction on n.

On $\mathcal{R}_{\mathrm{rad}}$, all radii are positive. The exact balancing formulas give

$$(s_i^y s_i^z)^3 = \frac{(\chi_i^x)^2}{\chi_i^y \chi_i^z}, \quad (s_i^x s_i^z)^3 = \frac{(\chi_i^y)^2}{\chi_i^x \chi_i^z}, \quad (s_i^x s_i^y)^3 = \frac{(\chi_i^z)^2}{\chi_i^x \chi_i^y}. \tag{A.17}$$

Every squared radius lies in $[n/2, 3n/2]$, while every product of two radii lies in the same interval $[n/2, 3n/2]$. Each ratio in (A.17) therefore lies in $[1/3, 3]$. Taking its positive cube root proves the three displayed bounds. This is a pointwise consequence on $\mathcal{R}_{\mathrm{rad}}$; it does not use or assert independence between the pair multipliers and the projected Gaussian coefficient vectors.

□

Lemma A.6 (rectangular isotropic Gaussian Khatri–Rao concentration)**.** *Under Assumption 3, let $Z = [z_1 \ \cdots \ z_k]$ and $W = [w_1 \ \cdots \ w_k]$ be independent $r \times k$ matrices with iid $\mathcal{N}(0,1)$ entries. Define*

$$K_0 = \frac{1}{r}[z_1 \otimes w_1 \ \cdots \ z_k \otimes w_k] \in \mathbb{R}^{r^2 \times k}.$$

If $r \geq 2^{40}$, then

$$\mathbb{P}\Big(\mathrm{spec}(K_0^\top K_0) \not\subset [1/2, 3/2]\Big) \leq r^{-26}. \tag{A.18}$$

Proof. Put $A = Z^\top Z$ and $L_r = 8\sqrt{\log r}$. First define

$$\mathcal{Z} := \left\{ \max_i \left| \frac{\|z_i\|_2^2}{r} - 1 \right| \leq \frac{1}{4}, \quad \|Z\|_{\mathrm{op}} \leq 3\sqrt{k} \right\}. \tag{A.19}$$

The same Laurent–Massart bound (Laurent and Massart, 2000, Lemma 1), with $d = r$, $\delta = 1/4$, followed by a union over the k columns, gives a first failure term $2ke^{-r/128}$. The rectangular Gaussian norm bound (Davidson and Szarek, 2001, Theorem II.13), applied to the $r \times k$ matrix Z with $u = L_r$, gives a second term r^{-32}. Since $r \geq 2^{40}$ implies $L_r \leq \sqrt{r} < \sqrt{k}$, its good-event threshold is at most $3\sqrt{k}$. The elementary numerical inequalities

$$2r^{5/4}e^{-r/128} \leq r^{-28}, \qquad r \geq 2^{40}, \tag{A.20}$$

and $r^{-32} \leq r^{-28}$ yield

$$\mathbb{P}(\mathcal{Z}^c) \leq r^{-27}. \tag{A.21}$$

On $\mathcal{Z}$,

$$\|A\|_{\mathrm{op}} \leq 9k, \qquad a_* := \max_i A_{ii} \leq \frac{5}{4}r, \qquad \frac{3}{4}I_k \preceq \frac{\mathrm{diag}(A)}{r} \preceq \frac{5}{4}I_k. \tag{A.22}$$

Write the rows of W as $w_{a\bullet}^\top$, $a \in [r]$, and let $D_a = \mathrm{diag}(w_{a1}, \ldots, w_{ak})$. The Hadamard/diagonal identity gives

$$K_0^\top K_0 = \frac{1}{r^2}\big(A \circ W^\top W\big) = \frac{1}{r^2}\sum_{a=1}^{r} D_a A D_a. \tag{A.23}$$

Introduce the entrywise truncation event

$$\mathcal{T}_W := \left\{ \max_{a,i} |w_{ai}| \leq L_r \right\}.$$

A scalar Gaussian union bound gives

$$\mathbb{P}(\mathcal{T}_W^c) \leq 2rke^{-L_r^2/2} \leq 2r^{9/4}r^{-32} \leq r^{-29}. \tag{A.24}$$

Because $\mathcal{T}_W$ is a product of coordinatewise truncation events, conditional on $\mathcal{T}_W$ all entries remain independent with the common law of a standard normal ξ conditioned on $|\xi| \leq L_r$. Put

$$\mu = \mathbb{E}\xi^2, \qquad \nu = \mathbb{E}\xi^4$$

under this conditional law. Symmetry gives $\mathbb{E}\xi = 0$. Since $L_r \geq 4$, integration of the Gaussian tail gives

$$\frac{7}{8} \leq \mu \leq 1, \qquad \nu \leq 6. \tag{A.25}$$

Indeed, $\mathbb{E}[g^2 \mathbf{1}_{\{|g|>L\}}] = 2(L\varphi(L) + \overline{\Phi}(L))$, so the conditional second-moment loss is below $1/8$ for $L \geq 4$; and the fourth numerator is at most $\mathbb{E}g^4 = 3$, while $\mathbb{P}(|g| \leq L) \geq 1/2$.

Fix any $Z \in \mathcal{Z}$, put $B = \operatorname{diag}(A)$, and, under the conditional law given $\mathcal{T}_W$, define the independent centered self-adjoint matrices

$$X_a = \frac{1}{r^2}(D_a A D_a - \mu B), \qquad a \in [r]. \tag{A.26}$$

The almost-sure truncation and (A.22) give

$$\|X_a\|_{\mathrm{op}} \le \frac{(L_r^2 + 1)\|A\|_{\mathrm{op}}}{r^2} \le \frac{18 L_r^2 k}{r^2} =: R_B. \tag{A.27}$$

For completeness, independence, symmetry, and (A.25) make $\mathbb{E}(D_a A D_a)^2$ diagonal. Its i-th diagonal entry is

$$\mu^2 (A^2)_{ii} + (\nu - \mu^2) A_{ii}^2.$$

After subtracting the square of the mean, the i-th diagonal entry of $\mathbb{E}(D_a A D_a - \mu B)^2$ is at most

$$(A^2)_{ii} + 6A_{ii}^2 \le \|A\|_{\mathrm{op}} A_{ii} + 6A_{ii}^2 \le 7\|A\|_{\mathrm{op}} a_*.$$

Consequently the matrix-Bernstein variance proxy satisfies

$$v_B := \left\| \sum_{a=1}^{r} \mathbb{E} X_a^2 \right\|_{\mathrm{op}} \le \frac{7\|A\|_{\mathrm{op}} a_*}{r^3} \le \frac{315}{4} \frac{k}{r^2} \le 79 \frac{k}{r^2}. \tag{A.28}$$

The self-adjoint matrix Bernstein inequality (Tropp, 2012, Theorem 6.1) states in the present two-sided form that independent centered self-adjoint $d \times d$ matrices with $\|X_a\|_{\mathrm{op}} \le R$ and $v = \|\sum_a \mathbb{E} X_a^2\|_{\mathrm{op}}$ obey

$$\mathbb{P}\left(\left\| \sum_a X_a \right\|_{\mathrm{op}} \ge t \right) \le 2d \exp\left(-\frac{t^2}{2v + (2Rt)/3} \right).$$

Here conditioning on $Z \in \mathcal{Z}$ and $\mathcal{T}_W$ leaves the matrices in the preceding display independent and centered, while the two preceding bounds give $R = R_B$ and $v = v_B$. Apply this inequality at $t = 1/8$. Using $L_r^2 = 64 \log r$, (A.27)–(A.28), and $k \le r^{5/4}$, its exponent is at least

$$\frac{(1/8)^2}{2v_B + (2R_B/8)/3} \ge \frac{r^2}{64(158 + 96 \log r)k} \ge \frac{r^{3/4}}{64(158 + 96 \log r)}. \tag{A.29}$$

For every $r \ge 2^{40}$,

$$\frac{r^{3/4}}{64(158 + 96 \log r)} \ge 30 \log r. \tag{A.30}$$

The inequality holds directly at 2^{40}, and the ratio of its left side to $\log r$ is increasing thereafter. Thus, uniformly for $Z \in \mathcal{Z}$,

$$\mathbb{P}\left(\left\| K_0^\top K_0 - \frac{\mu B}{r} \right\|_{\mathrm{op}} > \frac{1}{8} \;\middle|\; Z, \mathcal{T}_W \right) \le 2k r^{-30} \le r^{-28}. \tag{A.31}$$

On the complementary event in (A.31), (A.22) and (A.25) imply

$$\lambda_{\min}(K_0^\top K_0) \ge \frac{7}{8}\frac{3}{4} - \frac{1}{8} = \frac{17}{32} \ge \frac{1}{2}, \tag{A.32}$$

and

$$\lambda_{\max}(K_0^\top K_0) \le \frac{5}{4} + \frac{1}{8} = \frac{11}{8} \le \frac{3}{2}. \tag{A.33}$$

Finally, (A.21), (A.24), and (A.31), with no independence needed between their failures, give

$$\mathbb{P}\Big(\operatorname{spec}(K_0^\mathsf{T} K_0) \not\subset [1/2, 3/2]\Big) \le r^{-27} + r^{-29} + r^{-28} \le r^{-26}.$$

This proves the lemma.

□

Proposition A.3 (elliptic transfer with dependent balancing scalars)**.** *Under Proposition A.1, Lemmas A.3 and A.4, and Lemma A.5, condition on any realized triple in $\mathcal{E}_{\mathrm{cond}}$. For the three mode pairs define*

$$\begin{aligned}
K_0^{BC} &= \frac{1}{r}[z_1^B \otimes z_1^C \ \ \cdots \ \ z_k^B \otimes z_k^C], \quad D^{BC} = \operatorname{diag}(s_1^y s_1^z, \dots, s_k^y s_k^z),\\
K_0^{AC} &= \frac{1}{r}[z_1^A \otimes z_1^C \ \ \cdots \ \ z_k^A \otimes z_k^C], \quad D^{AC} = \operatorname{diag}(s_1^x s_1^z, \dots, s_k^x s_k^z),\\
K_0^{AB} &= \frac{1}{r}[z_1^A \otimes z_1^B \ \ \cdots \ \ z_k^A \otimes z_k^B], \quad D^{AB} = \operatorname{diag}(s_1^x s_1^y, \dots, s_k^x s_k^y).
\end{aligned} \tag{A.34}$$

If $\mathcal{R}_{\mathrm{rad}}$ holds and each corresponding isotropic Gram in (A.34) has spectrum in $[1/2, 3/2]$, then each exact normalized balanced pair Gram has spectrum in

$$\left[\frac{1}{2\, 3^{2/3}\kappa_1^4}, \frac{3^{5/3}}{2}\kappa_1^4\right]. \tag{A.35}$$

Proof. The normalized coefficient formula and the mixed-product identity for tensor products give

$$\begin{aligned}
\bar{K}_0^{\bar\beta\bar\gamma} &= (H_B \otimes H_C) K_0^{BC} D^{BC},\\
\bar{K}_0^{\bar\alpha\bar\gamma} &= (H_A \otimes H_C) K_0^{AC} D^{AC},\\
\bar{K}_0^{\bar\alpha\bar\beta} &= (H_A \otimes H_B) K_0^{AB} D^{AB}.
\end{aligned} \tag{A.36}$$

The factor $1/r$ in (A.34) is exactly the product of the two $1/\sqrt{r}$ normalized coefficient factors. Thus (A.36) uses the setting's normalized convention without a raw-scale substitution.

On $\mathcal{E}_{\mathrm{cond}}$, every Kronecker elliptic map in (A.36) has singular values in $[\kappa_1^{-2}, \kappa_1^2]$. On $\mathcal{R}_{\mathrm{rad}}$, Lemma A.5 gives

$$\sigma(D^{MN}) \subset [3^{-1/3}, 3^{1/3}]$$

for all three pairs. The assumed isotropic Gram event gives

$$\sigma(K_0^{MN}) \subset [2^{-1/2}, (3/2)^{1/2}].$$

Applying the checked product singular-value inequalities to (A.36) yields

$$\sigma_{\min}(\bar{K}_0^{MN}) \ge \kappa_1^{-2} 2^{-1/2} 3^{-1/3}, \qquad \|\bar{K}_0^{MN}\|_{\mathrm{op}} \le \kappa_1^2 (3/2)^{1/2} 3^{1/3}. \tag{A.37}$$

Squaring (A.37) gives exactly (A.35).

This transfer is deterministic on the intersection of the stated events. The diagonal D^{MN} can be arbitrarily dependent on K_0^{MN}: the pointwise singular-value comparison uses only its realized lower and upper diagonal entries. Hence no false scalar/Gaussian independence is introduced. The endpoint singular values of the H_M and endpoint pair multipliers are included in (A.37).

□

Proposition A.4 (simultaneous normalized pair-Gram event)**.** *Under Assumptions 2, 3, and 5, Proposition A.1, and Lemmas A.3 and A.4, define*

$$r_{0,\mathrm{gram}}(\kappa) := \max\left\{2^{40}, \left\lceil \left(2\,3^{2/3}\kappa_1^4\right)^{1/20}\right\rceil\right\}, \qquad \kappa_1 = 2\kappa^2. \tag{A.38}$$

Choose $C(\kappa, q) \geq 1$. Then, for every $r \geq r_{0,\mathrm{gram}}(\kappa)$, every $n \geq C(\kappa, q) r^4 \log r$, every $r < k \leq \lfloor r^{5/4} \rfloor$, and every realized factor triple in $\mathcal{E}_{\mathrm{cond}}$,

$$\mathbb{P}\left(\mathcal{E}_{\mathrm{gram}}^{\mathrm{norm}} \mid A, B, C\right) \geq 1 - r^{-20}. \tag{A.39}$$

Proof. Conditional on the realized factors, the three mode arrays $Z_M = [z_1^M \ \cdots \ z_k^M]$ are independent standard Gaussian arrays by the Gaussianization lemma. Therefore Lemma A.6 applies to each of the three pairs in (A.34), giving failure at most r^{-26} per pair. The pairs share mode arrays, but a union bound needs no independence. Together with Lemma A.5,

$$\mathbb{P}(\mathcal{R}_{\mathrm{rad}}^c \text{ or some isotropic pair event fails} \mid A, B, C) \leq r^{-28} + 3r^{-26} \leq 4r^{-26} \leq r^{-20}. \tag{A.40}$$

On the complementary event, Proposition A.3 places every normalized pair-Gram eigenvalue in the constant interval (A.35). The threshold (A.38) gives

$$r^{20} \geq 2\,3^{2/3}\kappa_1^4. \tag{A.41}$$

Consequently

$$\frac{1}{2\,3^{2/3}\kappa_1^4} \geq r^{-20}, \tag{A.42}$$

and, because

$$\frac{3^{5/3}}{2}\kappa_1^4 \leq 2\,3^{2/3}\kappa_1^4,$$

also

$$\frac{3^{5/3}}{2}\kappa_1^4 \leq r^{20}. \tag{A.43}$$

Thus all three exact normalized Grams lie in the setting-defined polynomial window on an event whose total conditional failure is bounded by (A.40). This is precisely $\mathcal{E}_{\mathrm{gram}}^{\mathrm{norm}}$, proving (A.39).

The proof is uniform in the realized factor triple because it uses that triple only through the common κ_1 singular interval. It is uniform in n above its lower threshold because the projected z_i^M law is exactly standard Gaussian for every n, while the full-radius failure in (A.16) decreases with n. The cases $k = r + 1$ and $k = \lfloor r^{5/4} \rfloor$ are both included, as are the endpoint elliptic and radial bounds.

□

Proof of the subsection conclusion. Lemma A.5 derives a full-radius event from the primitive ambient Gaussian initialization and proves uniform pointwise bounds for all three pairwise balancing diagonals. This is the only place where full radii enter, and their dependence on the projected coefficient Gaussians is retained rather than discarded.

Lemma A.6 proves both spectral sides for an $r^2 \times k$ isotropic Gaussian Khatri–Rao matrix. Its row decomposition (A.23) makes the conditional mean diagonal; entrywise Gaussian truncation supplies an almost-sure Bernstein envelope; (A.27)–(A.28) verify the matrix-Bernstein hypotheses; and (A.29)–(A.33) give the explicit lower and upper window with failure at most r^{-26}.

Proposition A.3 then uses the exact preceding formula

$$\bar{K}^{MN} = (H_M \otimes H_N) K_0^{MN} D^{MN}$$

with the setting's $1/r$ normalized scale. Pointwise singular-value comparisons transfer the isotropic window through the elliptic maps and the possibly dependent balancing diagonal, producing the constant interval (A.35) for each exact normalized Gram.

Finally, Proposition A.4 unions the single radial event and the three pair events, converts their total failure to at most r^{-20}, and verifies by (A.41)–(A.43) that the constant interval is contained in $[r^{-20}, r^{20}]$. These named results prove exactly all three normalized pair-Gram windows and nothing about the raw-scale bridge or $\mathcal{E}_{\text{size}}$.

$\square$

A.4 Balanced initialization radius

Lemma A.7 (raw Gaussian radius tail)**.** *Under Assumption 5, fix any realized factor triple (A, B, C) and any component-mode pair $(i, m) \in [k] \times \{x, y, z\}$. Then*

$$\mathbb{P}(\|\widetilde{m}_i\|_2 > 2 \mid A, B, C) \le \exp\left[-\frac{3 - \log 4}{2} n\right] \le e^{-n/2}.$$

Proof. Independence of initialization from smoothing implies that, conditionally on the realized factors, $g_i^m := \sqrt{n}\,\widetilde{m}_i$ remains a standard Gaussian vector in $\mathbb{R}^n$. For one standard normal scalar G, direct Gaussian integration gives

$$\mathbb{E}e^{\lambda G^2} = (1 - 2\lambda)^{-1/2}, \qquad 0 < \lambda < \frac{1}{2}.$$

Independence of the n coordinates therefore yields

$$\mathbb{E}e^{\lambda \|g_i^m\|_2^2} = (1 - 2\lambda)^{-n/2}.$$

Markov's inequality with $\lambda = 3/8$ gives

$$\begin{aligned}
\mathbb{P}(\|\widetilde{m}_i\|_2 > 2 \mid A, B, C) &= \mathbb{P}\big(\|g_i^m\|_2^2 > 4n \mid A, B, C\big) \\
&\le e^{-(3/8)4n}(1 - 3/4)^{-n/2} \\
&= e^{-3n/2}4^{n/2} = \exp\left[-\frac{3 - \log 4}{2} n\right].
\end{aligned}$$

Since $\log 4 < 2$, the exponent coefficient $(3 - \log 4)/2$ is larger than $1/2$, proving the second inequality. The calculation is uniform in the realized factors because their values do not enter the conditional Gaussian law.

$\square$

Proposition A.5 (simultaneous control of all raw initialization radii)**.** *Under Assumptions 2, 3, and 5, and Lemma A.7, define the event*

$$\mathcal{E}_{\text{raw,size}} := \left\{ \max_{i \in [k],\, m \in \{x,y,z\}} \|\widetilde{m}_i\|_2 \le 2 \right\}.$$

Choose the dimension-constant contribution $C_{\text{size}} := 1$ and the rank threshold $r_{\text{size}} := 3$. If

$$C(\kappa, q) \ge C_{\text{size}}, \qquad r \ge r_{\text{size}},$$

then, conditionally on every realized factor triple,

$$\mathbb{P}\big(\mathcal{E}^{\mathsf{c}}_{\text{raw,size}} \mid A, B, C\big) \le r^{-20}.$$

This conclusion holds throughout the full rank window, including the boundary $k = \lfloor r^{5/4} \rfloor$.

Proof. A union bound over the $3k$ raw columns and Lemma A.7 give

$$\mathbb{P}\big(\mathcal{E}^{\mathsf{c}}_{\text{raw,size}} \mid A, B, C\big) \le 3ke^{-n/2}.$$

No independence is needed for this union bound, although the columns are in fact independent. Assumptions 2 and 3, together with $C(\kappa, q) \ge 1$, imply

$$3ke^{-n/2} \le 3r^{5/4} \exp\left(-\frac{1}{2}r^4 \log r\right).$$

For $r \ge 3$,

$$\frac{r^4}{2} \ge \frac{81}{2} \ge \frac{85}{4} + \frac{\log 3}{\log r},$$

because $\log 3/\log r \le 1$. Multiplying by $\log r > 0$ and rearranging gives the explicit absorption inequality

$$\log 3 + \frac{5}{4}\log r - \frac{r^4}{2}\log r \le -20 \log r.$$

Exponentiation proves

$$3r^{5/4} \exp\left(-\frac{1}{2}r^4 \log r\right) \le r^{-20}.$$

The use of $k \le r^{5/4}$ is valid at $k = \lfloor r^{5/4} \rfloor$, so the maximal-rank boundary requires no separate loss. The lower inequality $r < k$ is part of the declared window but is not needed for this upper-tail estimate.

□

Proposition A.6 (geometric-mean transfer to balanced columns).
Under Assumption 5, Lemma A.3, and Proposition A.5, the deterministic inclusion

$$\mathcal{E}_{\text{raw,size}} \subseteq \mathcal{E}_{\text{size}}$$

holds. Hence, whenever $C(\kappa, q) \ge 1$ and $r \ge 3$, conditionally on every realized factor triple,

$$\mathbb{P}(\mathcal{E}^{\mathsf{c}}_{\text{size}} \mid A, B, C) \le r^{-20}.$$

Proof. Fix a component i on $\mathcal{E}_{\text{raw,size}}$. First suppose all three raw norms are positive. The balancing lemma gives

$$\|x_{i,0}\|_2 = \|y_{i,0}\|_2 = \|z_{i,0}\|_2 = \left(\|\widetilde{x}_i\|_2 \|\widetilde{y}_i\|_2 \|\widetilde{z}_i\|_2\right)^{1/3}.$$

Every raw norm on the right is at most 2, so

$$\left(\|\widetilde{x}_i\|_2 \|\widetilde{y}_i\|_2 \|\widetilde{z}_i\|_2\right)^{1/3} \le (2 \cdot 2 \cdot 2)^{1/3} = 2.$$

If at least one raw factor is exactly zero, the setting-defined zero branch leaves the entire raw triple unchanged. On $\mathcal{E}_{\text{raw,size}}$, each unchanged raw norm is already at most 2, including any nonzero factors in that triple. Thus the same balanced size conclusion holds on the zero branch without division by a raw norm and without removing a null event.

The argument applies to every $i \in [k]$, proving the event inclusion. Consequently

$$\mathbb{P}(\mathcal{E}^{\mathsf{c}}_{\text{size}} \mid A, B, C) \le \mathbb{P}\big(\mathcal{E}^{\mathsf{c}}_{\text{raw,size}} \mid A, B, C\big) \le r^{-20}$$

by Proposition A.5.

□

Proof of the subsection conclusion. Lemma A.7 derives from Assumption 5 the explicit conditional one-column bound $e^{-n/2}$. Proposition A.5 unions that bound over exactly the $3k$ raw initialization columns and uses the full admissible boundary $k \le \lfloor r^{5/4} \rfloor$, the explicit dimension contribution $C_{\mathrm{size}} = 1$, and the explicit threshold $r_{\mathrm{size}} = 3$ to obtain

$$\mathbb{P}\big(\mathcal{E}^{\mathsf{c}}_{\mathrm{raw,size}} \mid A, B, C\big) \le r^{-20}.$$

Proposition A.6, using the exact balancing law from Lemma A.3, proves on the positive-radius branch that each balanced norm is the geometric mean of three raw norms bounded by 2, and separately proves the same conclusion on the setting-defined zero-factor no-op branch. Therefore $\mathcal{E}_{\mathrm{raw,size}} \subseteq \mathcal{E}_{\mathrm{size}}$, and hence

$$\boxed{\mathbb{P}(\mathcal{E}^{\mathsf{c}}_{\mathrm{size}} \mid A, B, C) \le r^{-20}}$$

uniformly for every realized factor triple in $\mathcal{E}_{\mathrm{cond}}$, every $r \ge 3$, every $n \ge C(\kappa, q) r^4 \log r$ with $C(\kappa, q) \ge 1$, and every $r < k \le \lfloor r^{5/4} \rfloor$. This proves the claimed conditional failure bound for the balanced initial-size event. No Gram or path statement has been used or proved.

□

A.5 Raw and normalized tangent geometry

Lemma A.8 (Exact scaling of all three pair Grams)**.** *Under the setting definitions, Lemma A.4, and Proposition A.2, define*

$$\begin{aligned}
K^{\beta\gamma}_{\mathrm{raw}} &= [\, \beta_{1,0} \otimes \gamma_{1,0} \ \cdots \ \beta_{k,0} \otimes \gamma_{k,0} \,], \\
K^{\alpha\gamma}_{\mathrm{raw}} &= [\, \alpha_{1,0} \otimes \gamma_{1,0} \ \cdots \ \alpha_{k,0} \otimes \gamma_{k,0} \,], \\
K^{\alpha\beta}_{\mathrm{raw}} &= [\, \alpha_{1,0} \otimes \beta_{1,0} \ \cdots \ \alpha_{k,0} \otimes \beta_{k,0} \,],
\end{aligned}$$

and define the normalized matrices by replacing every coefficient vector by its barred version. For every pair $pq \in \{\beta\gamma, \alpha\gamma, \alpha\beta\}$, let $G^{pq}_{\mathrm{raw}} = (K^{pq}_{\mathrm{raw}})^{\mathsf{T}} K^{pq}_{\mathrm{raw}}$ and $G^{pq}_{\mathrm{norm}} = (K^{pq}_{\mathrm{norm}})^{\mathsf{T}} K^{pq}_{\mathrm{norm}}$. Then, pointwise on every branch,

$$K^{pq}_{\mathrm{raw}} = \frac{r}{n} K^{pq}_{\mathrm{norm}}, \qquad \boxed{G^{pq}_{\mathrm{raw}} = \left(\frac{r}{n}\right)^2 G^{pq}_{\mathrm{norm}}} \qquad \big(pq = \beta\gamma, \alpha\gamma, \alpha\beta\big).$$

Proof. Put $c = \sqrt{n/r} > 0$. The setting definitions give $\bar{\alpha}_{i,0} = c\alpha_{i,0}$, $\bar{\beta}_{i,0} = c\beta_{i,0}$, and $\bar{\gamma}_{i,0} = c\gamma_{i,0}$. Thus, for example,

$$\beta_{i,0} \otimes \gamma_{i,0} = c^{-2}(\bar{\beta}_{i,0} \otimes \bar{\gamma}_{i,0}) = \frac{r}{n}(\bar{\beta}_{i,0} \otimes \bar{\gamma}_{i,0}).$$

The identical calculation applies to the other two pairs, proving the matrix identity column by column. Taking transpose times the matrix gives

$$(K^{pq}_{\mathrm{raw}})^{\mathsf{T}} K^{pq}_{\mathrm{raw}} = \left(\frac{r}{n}\right)^2 (K^{pq}_{\mathrm{norm}})^{\mathsf{T}} K^{pq}_{\mathrm{norm}}.$$

No division by a coefficient vector occurs, so the calculation remains valid when a pair column is zero. This proves the lemma.

□

Proposition A.7 (Exact scaling and range of the tangent synthesis maps)**.** *Under the setting definitions, Lemma A.4, and Proposition A.2, let*

$$\mathcal{P} = \bigoplus_{i=1}^{k} (\mathbb{R}^r \oplus \mathbb{R}^r \oplus \mathbb{R}^r)$$

and write $p = ((u_i, v_i, w_i))_{i=1}^k \in \mathcal{P}$. *Define the raw and normalized tangent synthesis maps into* $\mathcal{H} = (\mathbb{R}^r)^{\otimes 3}$ *by*

$$\mathcal{T}_{\mathrm{raw}} p = \sum_{i=1}^{k} \big(u_i \otimes \beta_{i,0} \otimes \gamma_{i,0} + \alpha_{i,0} \otimes v_i \otimes \gamma_{i,0} + \alpha_{i,0} \otimes \beta_{i,0} \otimes w_i\big),$$
$$\mathcal{T}_{\mathrm{norm}} p = \sum_{i=1}^{k} \big(u_i \otimes \bar\beta_{i,0} \otimes \bar\gamma_{i,0} + \bar\alpha_{i,0} \otimes v_i \otimes \bar\gamma_{i,0} + \bar\alpha_{i,0} \otimes \bar\beta_{i,0} \otimes w_i\big).$$

Then, pointwise on every branch,

$$\boxed{\mathcal{T}_{\mathrm{raw}} = \frac{r}{n}\mathcal{T}_{\mathrm{norm}}}, \qquad \mathrm{range}(\mathcal{T}_{\mathrm{raw}}) = \mathscr{S}_0^{\mathrm{raw}}, \qquad \mathrm{range}(\mathcal{T}_{\mathrm{norm}}) = \mathscr{S}_0^{\mathrm{norm}}.$$

Consequently,

$$\boxed{\mathscr{S}_0^{\mathrm{raw}} = \mathscr{S}_0^{\mathrm{norm}} = \mathscr{S}_0}, \qquad \ker(\mathcal{T}_{\mathrm{raw}}) = \ker(\mathcal{T}_{\mathrm{norm}}).$$

Proof. With $c = \sqrt{n/r}$, every normalized tangent summand contains exactly two barred coefficient vectors and therefore equals $c^2 = n/r$ times the corresponding raw summand. Summing over components gives $\mathcal{T}_{\mathrm{norm}} = (n/r)\mathcal{T}_{\mathrm{raw}}$, equivalently the boxed map identity. The scalar r/n is nonzero.

Every image of $\mathcal{T}_{\mathrm{raw}}$ is a finite sum of generators in the definition of $\mathscr{S}_0^{\mathrm{raw}}$, so its range is contained in that span. Conversely, each displayed raw generator is obtained by setting exactly one of the variables u_i, v_i, w_i and setting all other variables to zero. Hence the range is the full displayed span. The normalized case is identical. Multiplication of a linear map by a nonzero scalar changes neither its range nor its kernel, proving both final equalities. Zero coefficient vectors merely remove zero generators from both maps and do not affect any argument. This proves the proposition. ∎

Lemma A.9 (Exact component gauge kernels and canonical gauge images). *Under the setting definitions, Lemma A.4, and Proposition A.2, let*

$$a_i = \alpha_{i,0}, \qquad b_i = \beta_{i,0}, \qquad c_i = \gamma_{i,0},$$

and let $\mathcal{T}_{\mathrm{raw},i}$ *denote the* i*-th summand map. Define the canonical gauge map and its image by*

$$\Gamma_i(\lambda, \mu) = \lambda(a_i, -b_i, 0) + \mu(a_i, 0, -c_i),$$
$$\mathfrak{n}_i = \mathrm{image}(\Gamma_i).$$

Then $\mathfrak{n}_i \subseteq \ker(\mathcal{T}_{\mathrm{raw},i})$ *on every branch. On the almost-sure branch supplied by Lemma A.4, where* $a_i, b_i, c_i \neq 0$,

$$\ker(\mathcal{T}_{\mathrm{raw},i}) = \{(\lambda a_i, \mu b_i, \nu c_i) : \lambda + \mu + \nu = 0\} = \mathfrak{n}_i, \qquad \dim \mathfrak{n}_i = 2.$$

On all branches the exact classification is:

Branch	*Local synthesis map*	*Local kernel*	*Kernel dimension*	$\dim \mathfrak{n}_i$
$a_i, b_i, c_i \neq 0$	*Full three-term map*	$\{(\lambda a_i, \mu b_i, \nu c_i) : \lambda + \mu + \nu = 0\}$	2	2
$a_i = 0,\ b_i, c_i \neq 0$	$u \otimes b_i \otimes c_i$	$\{0\} \oplus \mathbb{R}^r \oplus \mathbb{R}^r$	$2r$	2
$b_i = 0,\ a_i, c_i \neq 0$	$a_i \otimes v \otimes c_i$	$\mathbb{R}^r \oplus \{0\} \oplus \mathbb{R}^r$	$2r$	2
$c_i = 0,\ a_i, b_i \neq 0$	$a_i \otimes b_i \otimes w$	$\mathbb{R}^r \oplus \mathbb{R}^r \oplus \{0\}$	$2r$	2
$a_i \neq 0,\ b_i = c_i = 0$	0	$(\mathbb{R}^r)^3$	$3r$	1
$b_i \neq 0,\ a_i = c_i = 0$	0	$(\mathbb{R}^r)^3$	$3r$	1
$c_i \neq 0,\ a_i = b_i = 0$	0	$(\mathbb{R}^r)^3$	$3r$	1
$a_i = b_i = c_i = 0$	0	$(\mathbb{R}^r)^3$	$3r$	0

The normalized gauge map is $\Gamma_i^{\mathrm{norm}} = \sqrt{n/r}\,\Gamma_i$, so its image is the same $\mathfrak{n}_i$, and the normalized and raw local kernels agree.

Proof. Direct substitution gives

$$\begin{aligned}\mathcal{T}_{\mathrm{raw},i}(a_i, -b_i, 0) &= a_i \otimes b_i \otimes c_i - a_i \otimes b_i \otimes c_i = 0,\\ \mathcal{T}_{\mathrm{raw},i}(a_i, 0, -c_i) &= a_i \otimes b_i \otimes c_i - a_i \otimes b_i \otimes c_i = 0.\end{aligned}$$

This proves the all-branch inclusion without a nonvanishing assumption.

Now suppose $a_i, b_i, c_i \neq 0$ and $\mathcal{T}_{\mathrm{raw},i}(u, v, w) = 0$. Choose linear functionals b_i^*, c_i^* with $b_i^*(b_i) = c_i^*(c_i) = 1$, and let $P_{a_i^\perp}$ be orthogonal projection onto $a_i^\perp$. Applying $P_{a_i^\perp} \otimes b_i^* \otimes c_i^*$ gives $u = \lambda a_i$. Repeating in the other modes gives $v = \mu b_i$ and $w = \nu c_i$. Substitution yields

$$(\lambda + \mu + \nu) a_i \otimes b_i \otimes c_i = 0.$$

The rank-one tensor is nonzero, so $\lambda + \mu + \nu = 0$. Conversely, every such triple is killed by the map, and

$$(\lambda a_i, \mu b_i, \nu c_i) = -\mu(a_i, -b_i, 0) - \nu(a_i, 0, -c_i)$$

proves equality with $\mathfrak{n}_i$. The two generators are independent because all three coefficient vectors are nonzero.

If exactly one fixed vector is zero, the table's displayed one-term map survives and vanishes exactly when its free vector is zero; the two columns of Γ_i remain independent. If exactly two fixed vectors are zero, the local map is zero and the gauge image is the one-dimensional span of the surviving block vector. If all three vanish, both maps vanish. These observations prove the remaining table rows. The normalized conclusions follow from the common nonzero normalization scalar and Proposition A.7.

□

Proposition A.8 (Gauge quotient, unreduced range, and raw-target compatibility)**.** *Under the setting definitions, Lemma A.4, Proposition A.2, Proposition A.7, and Lemma A.9, embed each $\mathfrak{n}_i$ in the i-th parameter block and define*

$$\mathfrak{N}_{\mathrm{can}} = \bigoplus_{i=1}^{k} \mathfrak{n}_i \subseteq \mathcal{P}, \qquad \mathcal{Q}_{\mathrm{can}} = \mathcal{P}/\mathfrak{N}_{\mathrm{can}}.$$

Then both synthesis maps factor uniquely through the quotient map $\pi_{\mathrm{can}} : \mathcal{P} \to \mathcal{Q}_{\mathrm{can}}$:

$$\widetilde{\mathcal{T}}_{\mathrm{raw}}([p]) = \mathcal{T}_{\mathrm{raw}} p, \qquad \widetilde{\mathcal{T}}_{\mathrm{norm}}([p]) = \mathcal{T}_{\mathrm{norm}} p.$$

These induced maps satisfy

$$\begin{aligned}\mathrm{range}(\widetilde{\mathcal{T}}_{\mathrm{raw}}) &= \mathrm{range}(\mathcal{T}_{\mathrm{raw}}) = \mathscr{S}_0,\\ \mathrm{range}(\widetilde{\mathcal{T}}_{\mathrm{norm}}) &= \mathrm{range}(\mathcal{T}_{\mathrm{norm}}) = \mathscr{S}_0,\\ \widetilde{\mathcal{T}}_{\mathrm{raw}} &= \frac{r}{n}\widetilde{\mathcal{T}}_{\mathrm{norm}}.\end{aligned}$$

On the almost-sure all-nonzero branch, $\dim \mathfrak{N}_{\mathrm{can}} = 2k$, so this is exactly the two-gauge-directions-per-component quotient. On every zero branch it is the quotient by the branch-dependent, possibly lower-dimensional canonical gauge image. If $\mathcal{P} = \mathfrak{N}_{\mathrm{can}} \oplus \mathcal{C}_{\mathrm{can}}$, call a basis of $\mathcal{C}_{\mathrm{can}}$ a reduced parameter frame. Its synthesized columns have image range $\mathscr{S}_0$, but no injectivity or linear independence is claimed. Furthermore,

$$\boxed{\widehat{D}_0 \in \mathscr{S}_0}.$$

All these are raw coefficient-space statements: D_r is unchanged, $D_r - \widehat{D}_0$ is the raw coefficient residual in the Frobenius inner product on $\mathcal{H}$, and the physical loss remains the setting's $\|T - S(X,Y,Z)\|_F^2$ with no normalization.

Proof. Lemma A.9 gives $\mathfrak{n}_i \subseteq \ker(\mathcal{T}_{\mathrm{raw},i})$ for each component. Because the global map is the sum of its local maps, $\mathfrak{N}_{\mathrm{can}} \subseteq \ker(\mathcal{T}_{\mathrm{raw}})$. Proposition A.7 then gives $\mathfrak{N}_{\mathrm{can}} \subseteq \ker(\mathcal{T}_{\mathrm{norm}})$. Hence the two formulas on cosets are well defined: replacing p by $p+g$, $g \in \mathfrak{N}_{\mathrm{can}}$, does not change either image.

The quotient map π_{can} is surjective, so every quotient image is an unreduced image and every unreduced image is the image of the coset of its parameter. Therefore

$$\mathrm{range}(\widetilde{\mathcal{T}}_{\mathrm{raw}}) = \mathrm{range}(\mathcal{T}_{\mathrm{raw}}), \qquad \mathrm{range}(\widetilde{\mathcal{T}}_{\mathrm{norm}}) = \mathrm{range}(\mathcal{T}_{\mathrm{norm}}).$$

The remaining range and scaling identities follow from Proposition A.7. If $\mathcal{P} = \mathfrak{N}_{\mathrm{can}} \oplus \mathcal{C}_{\mathrm{can}}$, then for $p = g + c$, $\mathcal{T}_{\mathrm{raw}} p = \mathcal{T}_{\mathrm{raw}} c$; hence restriction to any such complement, and thus any reduced parameter frame, has the same image range as the unreduced generator frame. Additional kernel vectors may remain in $\mathcal{C}_{\mathrm{can}}$, so no basis or injectivity claim is made. An actual tangent-space basis would instead use the full common kernel $\ker(\mathcal{T}_{\mathrm{raw}}) = \ker(\mathcal{T}_{\mathrm{norm}})$.

For membership of the represented raw coefficient tensor, take $p_{\widehat{D}} = ((\alpha_{i,0}, 0, 0))_{i=1}^k$. Then

$$\mathcal{T}_{\mathrm{raw}} p_{\widehat{D}} = \sum_{i=1}^k \alpha_{i,0} \otimes \beta_{i,0} \otimes \gamma_{i,0} = \widehat{D}_0,$$

so $\widehat{D}_0 \in \mathrm{range}(\mathcal{T}_{\mathrm{raw}}) = \mathscr{S}_0$, including on every zero branch. Neither the quotient nor the coefficient normalization acts on D_r, and no map here acts on the physical tensor space. This proves every raw-target and metric convention in the statement.

The global kernel of $\mathcal{T}_{\mathrm{raw}}$ can contain additional cross-component linear dependencies. No equality $\ker(\mathcal{T}_{\mathrm{raw}}) = \mathfrak{N}_{\mathrm{can}}$ is claimed or needed. On the almost-sure nonzero branch the canonical image removes exactly two local gauge directions per component; on zero branches its dimension is the one in Lemma A.9. In every case, inclusion in the full kernel is the precise condition that preserves the range.

□

Proof of the subsection conclusion. Lemma A.8 applies the setting normalization column by column to all three Khatri–Rao pair matrices and proves exactly

$$G^{pq}_{\mathrm{raw}} = \left(\frac{r}{n}\right)^2 G^{pq}_{\mathrm{norm}}, \qquad pq \in \{\beta\gamma, \alpha\gamma, \alpha\beta\}.$$

It makes no claim about eigenvalues, concentration, or a Gram event.

Proposition A.7 defines the unreduced raw and normalized tangent synthesis maps and proves the exact nonzero scalar identity between them. Their ranges are, by direct two-sided comparison with the setting generators, precisely $\mathscr{S}_0^{\mathrm{raw}}$ and $\mathscr{S}_0^{\mathrm{norm}}$. This yields $\mathscr{S}_0^{\mathrm{raw}} = \mathscr{S}_0^{\mathrm{norm}} = \mathscr{S}_0$ without a surrogate span or metric change.

Lemma A.9 identifies the exact two-dimensional local scaling kernel on the previously established almost-sure nonzero branch and gives the exact kernel and canonical gauge-image dimensions on every zero branch. Proposition A.8 then factors both synthesis maps through the canonical-image quotient. On the nonzero branch this removes exactly two gauge directions per component; on every branch the quotient, unreduced generator frame, and complement-based reduced parameter frame have the same image range $\mathscr{S}_0$, with no injectivity claim. The same proposition also proves the exact membership $\widehat{D}_0 \in \mathscr{S}_0$.

Thus the range and membership claims are proved. In particular, the exact raw, normalized, and quotient ranges all equal $\mathscr{S}_0$, while $\widehat{D}_0 \in \mathscr{S}_0$ and the raw target, residual, and Frobenius conventions remain unchanged. No probability, Gram concentration, full-rank Gaussian-array, Haar, projection-energy, or deficit conclusion is asserted.

□

A.6 Product-Haar orbit factorization

For each $M \in \{A, B, C\}$, define the standard-Gaussian mode array

$$Z_M = [z_1^M \ \cdots \ z_k^M] \in \mathbb{R}^{r\times k},$$

using the vectors from Lemma A.4. The arrays Z_A, Z_B, Z_C correspond respectively to modes a, b, c, and we use the accepted mode abbreviations

$$H_a := H_A, \qquad H_b := H_B, \qquad H_c := H_C.$$

Lemma A.10 (Gaussian block QR separates orientation from the full rotated shape)**.** *Under Assumptions 5 and 3, where the latter gives $r < k$, hence $k \geq r$, and Lemma A.4, let $Z \in \mathbb{R}^{r\times k}$ have iid $\mathcal{N}(0,1)$ entries. Write*

$$Z = [G\ W], \qquad G \in \mathbb{R}^{r\times r}, \qquad W \in \mathbb{R}^{r\times(k-r)}.$$

There are Borel functions $O(Z) \in O(r)$ and $R(Z) \in \mathbb{R}^{r\times k}$ such that

$$Z = O(Z)R(Z) \tag{A.44}$$

for every Z, and, under the Gaussian law,

$$O(Z) \sim \mathrm{Haar}(O(r)), \qquad O(Z) \perp\!\!\!\perp R(Z). \tag{A.45}$$

On the full-probability event $\Omega_{\mathrm{qr}} = \{\det G \neq 0\}$, if

$$G = O_0 R_0$$

is the unique QR decomposition with R_0 upper triangular and strictly positive diagonal, then the construction is explicitly

$$O(Z) = O_0, \qquad Y = O_0^\mathsf{T} W, \qquad R(Z) = [R_0\ Y]. \tag{A.46}$$

Thus the orientation is independent not only of the triangular factor but of the complete shape containing all rotated remaining Gaussian columns.

Proof. The determinant is a nonzero polynomial in the entries of G, and the Gaussian law has a density, so

$$\mathbb{P}(\Omega_{\mathrm{qr}}^c) = 0. \tag{A.47}$$

On Ω_{qr}, positive-diagonal QR is unique. Moreover, R_0 is the unique upper-triangular positive-diagonal Cholesky factor of $G^\mathsf{T} G$, and

$$O_0 = G R_0^{-1}. \tag{A.48}$$

The Cholesky map is continuous on the positive-definite cone, so O_0, R_0 are Borel functions of G there. On Ω_{qr}^c, set $O(Z) = I_r$, and on all branches set

$$R(Z) = O(Z)^\mathsf{T} Z.$$

This is a Borel extension and proves equation (A.44) pointwise. On Ω_{qr} it agrees with equation (A.46).

For a deterministic $U \in O(r)$, the positive-diagonal QR decomposition of UG is

$$UG = (UO_0)R_0. \tag{A.49}$$

Because $UG \overset{d}{=} G$, equation (A.49) implies

$$(O_0, R_0) \overset{d}{=} (UO_0, R_0). \tag{A.50}$$

For every Borel set B in the triangular-factor space, the finite measure

$$\mu_B(A) = \mathbb{P}(O_0 \in A, R_0 \in B)$$

on $O(r)$ is left invariant by equation (A.50). Haar uniqueness therefore gives

$$\mu_B(A) = \mathrm{Haar}_{O(r)}(A)\,\mathbb{P}(R_0 \in B), \tag{A.51}$$

so O_0 is Haar and independent of R_0. Here the invoked compact-group fact is uniqueness of normalized Haar measure: every finite left-invariant Borel measure is a scalar multiple of it (Folland, 1999, Chapter 2). Applying that statement to each μ_B gives exactly the displayed product factorization, including the cases $\mathbb{P}(R_0 \in B) = 0$.

It remains to include the unused columns in the shape. The Gaussian block W is independent of G, hence of (O_0, R_0). Conditional on $(O_0, R_0) = (o, t)$, orthogonal invariance of every column of W gives

$$o^\top W \overset{d}{=} W, \tag{A.52}$$

and this conditional law does not depend on o or t. More explicitly, for bounded Borel f and g, define $h(t) = \mathbb{E}[g(t, W)]$. Then

$$\begin{aligned} \mathbb{E}[f(O_0)g(R_0, O_0^\top W)] &= \mathbb{E}[f(O_0)h(R_0)] \\ &= \mathbb{E}[f(O_0)]\,\mathbb{E}[h(R_0)] \\ &= \mathbb{E}[f(O_0)]\,\mathbb{E}[g(R_0, O_0^\top W)]. \end{aligned} \tag{A.53}$$

Thus O_0 is independent of (R_0, Y), proving equations (A.45)–(A.46). The measurable extension changes the variables only on the null event in equation (A.47), so the Haar and independence laws remain exact. Since invertibility of G implies full row rank of Z, every rank-deficient branch is contained in the same null event. The argument also covers $k = r + 1$, where W has exactly one column.

□

Proposition A.9 (Reflection-bit absorption gives an independent Haar orientation in $SO(r)$)**.** *Under Lemma A.10, fix*

$$J = \mathrm{diag}(-1, 1, \ldots, 1) \in O(r), \qquad \det J = -1,$$

and write $Z = O_0 R$ for the lemma's factorization. Define the measurable determinant bit, orientation, and reflected shape by

$$\varepsilon = \mathbf{1}_{\{\det O_0 = -1\}}, \qquad Q = O_0 J^\varepsilon, \qquad \widetilde{R} = J^\varepsilon R. \tag{A.54}$$

Then

$$Q \in SO(r), \qquad Z = Q\widetilde{R}, \tag{A.55}$$

Q is Haar on $SO(r)$, and

$$Q \perp\!\!\!\perp \widetilde{R}. \tag{A.56}$$

The bit ε is stored entirely in $\widetilde{R}$.

Proof. Since $J^2 = I_r$, equation (A.54) gives

$$\det Q = (\det O_0)(-1)^{\varepsilon} = 1, \qquad Q\widetilde{R} = O_0 J^{\varepsilon} J^{\varepsilon} R = O_0 R = Z,$$

which proves equation (A.55) on every branch.

Normalized Haar measure on $O(r)$ assigns mass $1/2$ to each determinant component because right multiplication by J interchanges them. On the component $\det O_0 = 1$, $Q = O_0$; on the component $\det O_0 = -1$, right multiplication by J maps that component bijectively and measure preservingly onto $SO(r)$. Hence

$$Q \sim \mathrm{Haar}(SO(r)), \qquad Q \perp\!\!\!\perp \varepsilon. \tag{A.57}$$

Lemma A.10 gives $O_0 \perp\!\!\!\perp R$. The measurable bijection $O_0 \leftrightarrow (Q, \varepsilon)$ therefore gives $(Q, \varepsilon) \perp\!\!\!\perp R$. Combining this with equation (A.57), for bounded Borel f, g,

$$\begin{aligned} \mathbb{E}[f(Q)g(J^{\varepsilon}R)] &= \mathbb{E}[g(J^{\varepsilon}R)\mathbb{E}[f(Q) \mid \varepsilon, R]] \\ &= \mathbb{E}[f(Q)]\,\mathbb{E}[g(J^{\varepsilon}R)], \end{aligned}$$

which proves equation (A.56). Thus absorbing the reflection does not leave a determinant bit correlated with the exposed $SO(r)$ orientation.

□

Proposition A.10 (Product-Haar orientation–shape factorization)**.**
Under Assumptions 5 and 3, where the latter gives $r < k$, hence $k \geq r$, Lemma A.4, and Proposition A.9, there are measurable factorizations

$$Z_A = Q_a R_A, \qquad Z_B = Q_b R_B, \qquad Z_C = Q_c R_C, \tag{A.58}$$

such that, conditional on the realized factors,

$$(Q_a, Q_b, Q_c) \sim \mathrm{Haar}(SO(r))^{\otimes 3}, \qquad (Q_a, Q_b, Q_c) \perp\!\!\!\perp (R_A, R_B, R_C). \tag{A.59}$$

Writing r_i^M for column i of R_M, define

$$E = \mathrm{span}_{i \in [k]}\{u \otimes r_i^B \otimes r_i^C, r_i^A \otimes v \otimes r_i^C, r_i^A \otimes r_i^B \otimes w : \quad u, v, w \in \mathbb{R}^r\}. \tag{A.60}$$

Then E is a measurable random subspace,

$$(Q_a, Q_b, Q_c) \perp\!\!\!\perp E, \qquad \boxed{\dim(E) \leq 3kr}. \tag{A.61}$$

Proof. Conditional on the realized factor triple, the three arrays Z_A, Z_B, Z_C are independent standard Gaussian arrays by the Gaussianization lemma. Apply Proposition A.9 separately to each array. Each pair (Q_m, R_M) is a measurable function of its own mode array, so the three pairs are independent. Within each pair the orientation is Haar and independent of its shape. The joint law consequently factors as

$$\bigotimes_{m \in \{a,b,c\}} \left(\mathrm{Haar}_{SO(r)} \otimes \mathcal{L}(R_M)\right),$$

which is exactly equation (A.59), not merely pairwise independence.

For measurability of equation (A.60), define the internal synthesis map

$$\begin{aligned} \mathcal{T}_R((u_i, v_i, w_i)_{i=1}^k) = \sum_{i=1}^{k} (&u_i \otimes r_i^B \otimes r_i^C + r_i^A \otimes v_i \otimes r_i^C \\ &+ r_i^A \otimes r_i^B \otimes w_i). \end{aligned} \tag{A.62}$$

In standard bases, the matrix V_R of $\mathcal{T}_R$ has entries that are polynomial in the entries of R_A, R_B, R_C, and $E = \mathrm{range}(V_R)$. Its orthogonal projector is the Borel function

$$P_E = V_R V_R^\dagger, \qquad V_R^\dagger = \lim_{j\to\infty} (V_R^\top V_R + j^{-1} I)^{-1} V_R^\top, \tag{A.63}$$

where the limit follows directly from a singular-value decomposition. Thus the random subspace is measurable. Since it is a function only of the three shapes, equation (A.59) proves the independence in equation (A.61).

Finally, the domain of $\mathcal{T}_R$ has dimension $3kr$, so

$$\dim(E) = \mathrm{rank}(\mathcal{T}_R) \le 3kr.$$

This remains true if some shape column vanishes or if cross-component dependencies enlarge the synthesis kernel. It makes no injectivity or exact dimension claim. At $k = r + 1$, each QR shape contains one rotated remaining column, and the same construction and bound apply without modification.

□

Proposition A.11 (Exact raw tangent equivariance)**.** *Under Lemma A.4, Proposition A.2, Propositions A.7 and A.8, and Proposition A.10, define*

$$L = H_a \otimes H_b \otimes H_c, \qquad Q = Q_a \otimes Q_b \otimes Q_c. \tag{A.64}$$

Then L and Q are invertible coefficient-tensor operators and

$$\boxed{\mathscr{S}_0 = (H_a \otimes H_b \otimes H_c)(Q_a \otimes Q_b \otimes Q_c)E = LQE.} \tag{A.65}$$

This is equality with the exact raw tangent range from the setting and the quotient convention.

Proof. By equation (A.58), for every component and mode,

$$z_i^A = Q_a r_i^A, \qquad z_i^B = Q_b r_i^B, \qquad z_i^C = Q_c r_i^C. \tag{A.66}$$

Let E_i^a, E_i^b, E_i^c denote the three component subspaces displayed in equation (A.60). For the first one,

$$\begin{aligned} LQE_i^a &= \{H_a Q_a u \otimes H_b Q_b r_i^B \otimes H_c Q_c r_i^C : u \in \mathbb{R}^r\} \\ &= \{u' \otimes H_b z_i^B \otimes H_c z_i^C : u' \in \mathbb{R}^r\}, \end{aligned} \tag{A.67}$$

because $H_a Q_a$ is invertible. The same argument in the other two free modes gives

$$\begin{aligned} LQE_i^b &= \{H_a z_i^A \otimes v' \otimes H_c z_i^C : v' \in \mathbb{R}^r\}, \\ LQE_i^c &= \{H_a z_i^A \otimes H_b z_i^B \otimes w' : w' \in \mathbb{R}^r\}. \end{aligned} \tag{A.68}$$

Taking the span over i and comparing equations (A.67)–(A.68) with the exact raw-range identity from Proposition A.2 gives

$$\begin{aligned} \mathscr{S}_0 = \mathrm{span}_{i\in[k]}\{&u \otimes H_b z_i^B \otimes H_c z_i^C, H_a z_i^A \otimes v \otimes H_c z_i^C, \\ &H_a z_i^A \otimes H_b z_i^B \otimes w : u, v, w \in \mathbb{R}^r\}. \end{aligned} \tag{A.69}$$

The three ranges in (A.67)–(A.68) are precisely the three displayed generator families, so their span proves equation (A.65).

For completeness, the actual raw first-mode tangent block contains the factor $s_i^y s_i^z/n$, while its normalized counterpart contains $s_i^y s_i^z/r$; the analogous factors occur in the other two modes. The balancing-invariance proposition proves that every such factor is nonzero, including its exact scalar convention on the no-op zero branch, so each multiplies a whole block by a bijective scalar. Thus no balancing scalar, normalization scalar, or independence assertion about those scalars is hidden in equation (A.65). Proposition A.8 then identifies this range with the raw, normalized, and quotient ranges. The fixed coefficient target D_r, $\widehat{D}_0$, and every physical-space object are not acted on or redefined in this proof.

□

Proof of the subsection conclusion. Lemma A.10 applies positive-diagonal QR to the first r columns of each standard Gaussian mode array. It proves explicitly that the QR orientation is Haar on $O(r)$ and independent of the complete rotated shape $[R_0 \ O_0^\top W]$, not only of the triangular factor. The singular first-block and rank-deficient branches are null; the piecewise definition still gives a measurable exact factorization on those branches.

Proposition A.9 writes each Haar $O(r)$ orientation as $Q_m J^{\varepsilon_m}$, with Q_m Haar on $SO(r)$, and replaces the shape by $J^{\varepsilon_m} R_M$. It proves both the exact identity $Z_M = Q_m R_M$ after this replacement and independence of the exposed orientation from the reflected shape. Therefore no untracked determinant bit remains outside the internal shape.

Proposition A.10 uses independence of the three Gaussian mode arrays to obtain the product law $\mathrm{Haar}(SO(r))^{\otimes 3}$, jointly independent of all three shapes. It defines the measurable internal tangent subspace E from those shapes and proves

$$(Q_a, Q_b, Q_c) \perp\!\!\!\perp E, \qquad \dim(E) \le 3kr.$$

The proof explicitly includes the boundary $k = r + 1$, null rank branch, and possible rank loss of the internal synthesis map.

Finally, the exact identities $z_i^M = Q_m r_i^M$, together with Propositions A.2 and A.8, give the blockwise identity

$$\mathscr{S}_0 = (H_a \otimes H_b \otimes H_c)(Q_a \otimes Q_b \otimes Q_c)E.$$

These results establish the claimed product-Haar factorization. No projection of D_r, Haar mean, concentration inequality, leverage bound, or tangent-deficit conclusion is asserted.

□

A.7 Fixed-target product-Haar projection

For a fixed subspace $E \subset \mathcal{H} = (\mathbb{R}^r)^{\otimes 3}$, a fixed nonzero tensor $X \in \mathcal{H} \setminus \{0\}$, and $Q = Q_a \otimes Q_b \otimes Q_c$, define

$$h(Q; E, X) := \frac{\|P_{QE} X\|_F^2}{\|X\|_F^2}.$$

The denominator is therefore strictly positive; no $X = 0$ case is included. If $d = \dim(E) = 0$, then $QE = \{0\}$ and $h(Q; E, X) = 0$.

Lemma A.11 (Sequential $SO(r)$ twirling gives the exact projection mean)**.** *Under Assumption 2, using only $r \ge 3$, let $E \subset \mathcal{H} = (\mathbb{R}^r)^{\otimes 3}$ be a fixed subspace of dimension d, let $X \in \mathcal{H} \setminus \{0\}$, and let Q_a, Q_b, Q_c be independent Haar matrices in $SO(r)$. Then*

$$\mathbb{E} \frac{\|P_{(Q_a \otimes Q_b \otimes Q_c)E} X\|_F^2}{\|X\|_F^2} = \frac{d}{r^3}. \tag{A.70}$$

Proof. We first prove the one-mode identity in the exact real representation used here. Let $V = \mathbb{R}^r$, let W be any finite-dimensional real inner-product space, and let $A \in \mathrm{End}(V \otimes W)$. Define

$$\mathcal{T}_V(A) = \int_{SO(r)} (U \otimes I_W) A (U^\top \otimes I_W)\, dU. \tag{A.71}$$

Haar invariance shows that $\mathcal{T}_V(A)$ commutes with $S \otimes I_W$ for every $S \in SO(r)$.

For completeness, the commutant of the natural real $SO(r)$-action is scalar when $r \geq 3$. Indeed, if $C \in \mathrm{End}(V)$ commutes with $SO(r)$, then for every unit vector u, the vector Cu is fixed by the subgroup that fixes u and rotates $u^\perp$. The fixed space of that subgroup is $\mathrm{span}\{u\}$, so every unit vector is an eigenvector of C. Applying linearity to two independent vectors and their sum shows that all eigenvalues coincide; hence C is scalar. Taking matrix coefficients of $\mathcal{T}_V(A)$ between arbitrary vectors of W now gives

$$\mathcal{T}_V(A) = I_V \otimes B$$

for some $B \in \mathrm{End}(W)$. Partial trace over V is unchanged by the conjugations in equation (A.71), whereas $\mathrm{Tr}_V(I_V \otimes B) = rB$. Therefore

$$\boxed{\mathcal{T}_V(A) = \frac{I_V}{r} \otimes \mathrm{Tr}_V(A).} \tag{A.72}$$

Let P_E be the orthogonal projector onto E. Since the tensor product of orthogonal maps is orthogonal,

$$P_{(Q_a \otimes Q_b \otimes Q_c)E} = (Q_a \otimes Q_b \otimes Q_c) P_E (Q_a^\top \otimes Q_b^\top \otimes Q_c^\top). \tag{A.73}$$

Apply equation (A.72) successively in modes a, b, c. Independence permits the three iterated Haar integrals, and the commuting mode actions give

$$\begin{aligned} \mathbb{E}[Q P_E Q^\top] &= \frac{I_r}{r} \otimes \frac{I_r}{r} \otimes \frac{I_r}{r} \,\mathrm{Tr}(P_E) \\ &= \frac{d}{r^3} I_{\mathcal{H}}. \end{aligned} \tag{A.74}$$

Thus

$$\mathbb{E}h = \frac{\langle X, \mathbb{E}[Q P_E Q^\top] X\rangle}{\|X\|_F^2} = \frac{d}{r^3},$$

which proves equation (A.70).

If $d = 0$, then $P_E = 0$ and $h = 0$ identically, in agreement with the formula. The proof uses only $\mathrm{Tr}(P_E) = d$, so it remains valid at every actual endpoint, including $d = 3kr$ whenever a subspace of that dimension is supplied. No separate inequality $3kr < r^3$ is assumed; the existence of $E \subset \mathcal{H}$ already enforces $d \leq r^3$.

□

Lemma A.12 (Metric-exact Lipschitz bound for product rotations)**.** *Fix $E \subset \mathcal{H}$ and $X \in \mathcal{H} \setminus \{0\}$. Equip each $SO(r)$ factor with the bi-invariant metric whose identity inner product is*

$$\langle K, L\rangle_F = -\,\mathrm{tr}(KL), \qquad K, L \in \mathfrak{so}(r),$$

and equip $SO(r)^3$ with the product metric. Then

$$\boxed{\mathrm{Lip}(h) \leq 2\sqrt{3}}. \tag{A.75}$$

Equivalently, a product tangent vector with skew generators (K_a, K_b, K_c) *has squared speed*

$$\sum_{m \in \{a,b,c\}} \|K_m\|_F^2,$$

with no normalization factor, and the directional derivative of h *is at most* $2\sqrt{3}$ *times that speed.*

Proof. By left invariance, a product geodesic through (Q_a, Q_b, Q_c) can be written

$$Q_m(s) = e^{sK_m} Q_m, \qquad K_m \in \mathfrak{so}(r),$$

and its speed is exactly

$$\left(\|K_a\|_F^2 + \|K_b\|_F^2 + \|K_c\|_F^2\right)^{1/2}. \tag{A.76}$$

The three tensor-mode generators commute, so the induced tensor action is

$$\begin{aligned} Q(s) &= e^{s\mathcal{K}} Q, \\ \mathcal{K} &= K_a \otimes I_r \otimes I_r + I_r \otimes K_b \otimes I_r + I_r \otimes I_r \otimes K_c. \end{aligned} \tag{A.77}$$

This gives the explicit tensor-generator estimate

$$\begin{aligned} \|\mathcal{K}\|_{\mathrm{op}} &\leq \|K_a\|_{\mathrm{op}} + \|K_b\|_{\mathrm{op}} + \|K_c\|_{\mathrm{op}} \\ &\leq \|K_a\|_F + \|K_b\|_F + \|K_c\|_F \\ &\leq \sqrt{3}\left(\|K_a\|_F^2 + \|K_b\|_F^2 + \|K_c\|_F^2\right)^{1/2}. \end{aligned} \tag{A.78}$$

Write $P = P_{QE}$. Along the geodesic,

$$P(s) = e^{s\mathcal{K}} P e^{-s\mathcal{K}}, \qquad P'(0) = [\mathcal{K}, P]. \tag{A.79}$$

Consequently,

$$\begin{aligned} |h'(0)| &= \frac{|\langle X, [\mathcal{K}, P]X\rangle|}{\|X\|_F^2} \\ &\leq \|[\mathcal{K}, P]\|_{\mathrm{op}} \leq 2\|\mathcal{K}\|_{\mathrm{op}} \\ &\leq 2\sqrt{3}\left(\|K_a\|_F^2 + \|K_b\|_F^2 + \|K_c\|_F^2\right)^{1/2}. \end{aligned} \tag{A.80}$$

Here $\|P\|_{\mathrm{op}} \leq 1$, including the case $E = \{0\}$. The bound holds at every point and along every geodesic. Integrating equation (A.80) along a minimizing geodesic proves equation (A.75) for exactly the metric in the statement.

□

Proposition A.12 (Ricci–log-Sobolev–Herbst tail in the unnormalized product metric)**.** *Under Assumption 2, using only* $r \geq 3$*, and under the hypotheses of Lemma A.11, one has for every* $t > 0$

$$\mathbb{P}(h - \mathbb{E}h \geq t) \leq \exp\left(-\frac{(r-2)t^2}{96}\right) \leq \exp\left(-\frac{rt^2}{288}\right). \tag{A.81}$$

Combining this with Lemma A.11, for $0 < t \leq 1$,

$$\mathbb{P}\left(h \geq \frac{d}{r^3} + t\right) \leq 8\exp(-c_H r t^2), \qquad \boxed{c_H = \frac{1}{288}}. \tag{A.82}$$

Proof. We first verify the metric normalization rather than importing a scale-free concentration slogan. On $\mathfrak{so}(r)$, use the inner product stated in Lemma A.12. The matrices

$$F_{ij} = \frac{E_{ij} - E_{ji}}{\sqrt{2}}, \qquad 1 \le i < j \le r,$$

form an orthonormal basis. The Koszul formula for left-invariant fields and the Ad-invariance of that inner product give the standard bi-invariant compact-group connection and curvature formulas (Besse, 1987, Chapter 7), here written in our curvature-sign convention,

$$\nabla_K L = \frac{1}{2}[K, L], \qquad R(K, L)M = -\frac{1}{4}[[K, L], M]. \tag{A.83}$$

Hence

$$\mathrm{Ric}(K, K) = \frac{1}{4}\sum_{i<j} \|[K, F_{ij}]\|_F^2 = -\frac{1}{4}\,\mathrm{tr}(\mathrm{ad}_K^2). \tag{A.84}$$

A direct summation in the displayed matrix-unit basis gives, for $K, L \in \mathfrak{so}(r)$,

$$\mathrm{tr}(\mathrm{ad}_K\,\mathrm{ad}_L) = (r-2)\,\mathrm{tr}(KL). \tag{A.85}$$

Since $\mathrm{tr}(K^2) = -\|K\|_F^2$, equations (A.84)–(A.85) yield the exact identity

$$\mathrm{Ric}_{SO(r)}(K, K) = \frac{r-2}{4}\|K\|_F^2. \tag{A.86}$$

This calculation is tied to the unnormalized basis F_{ij}; replacing the metric by $-\,\mathrm{tr}(KL)/2$ or by a dimension-normalized metric would change equation (A.86), but no such replacement is made here.

For the product metric and $\mathbf{K} = (K_a, K_b, K_c)$, Ricci curvature adds over factors:

$$\begin{aligned} \mathrm{Ric}_{SO(r)^3}(\mathbf{K}, \mathbf{K}) &= \sum_{m\in\{a,b,c\}} \frac{r-2}{4}\|K_m\|_F^2 \\ &= \rho\|\mathbf{K}\|_g^2, \qquad \rho = \frac{r-2}{4}. \end{aligned} \tag{A.87}$$

In current notation, the Bakry–Emery criterion says that on a complete connected Riemannian manifold with normalized volume μ and $\mathrm{Ric} \ge \rho g$, $\rho > 0$,

$$\mathrm{Ent}_\mu(f^2) \le \frac{2}{\rho}\int \|\nabla f\|_g^2\, d\mu$$

for smooth f, where

$$\mathrm{Ent}_\mu(Y) := \int Y \log Y\, d\mu - \left(\int Y\, d\mu\right)\log\left(\int Y\, d\mu\right).$$

This is the Bakry–Emery log-Sobolev implication (Bakry et al., 2014, Chapter 5). Compactness supplies completeness, $SO(r)^3$ is connected for $r \ge 3$, and the preceding calculation gives $\rho = (r-2)/4$, so every hypothesis is discharged for normalized product Haar measure. Let

$$Y = h - \mathbb{E}h, \qquad \psi(\lambda) = \log \mathbb{E}e^{\lambda Y}, \qquad \lambda > 0.$$

Apply this log-Sobolev inequality to $f = e^{\lambda Y/2}$. By Lemma A.12, $\|\nabla h\|_g \le L$ with $L = 2\sqrt{3}$, so

$$\begin{aligned} \mathrm{Ent}(e^{\lambda Y}) &\le \frac{2}{\rho}\frac{\lambda^2}{4}\mathbb{E}[e^{\lambda Y}\|\nabla h\|_g^2] \\ &\le \frac{\lambda^2 L^2}{2\rho}\mathbb{E}e^{\lambda Y}. \end{aligned} \tag{A.88}$$

On the other hand,

$$\frac{\mathrm{Ent}(e^{\lambda Y})}{\mathbb{E}e^{\lambda Y}} = \lambda\psi'(\lambda) - \psi(\lambda). \tag{A.89}$$

Dividing equations (A.88)–(A.89) by λ^2, integrating the derivative of $\psi(\lambda)/\lambda$ from 0 to λ, and using $\mathbb{E}Y = 0$, gives

$$\psi(\lambda) \le \frac{\lambda^2 L^2}{2\rho}. \tag{A.90}$$

Chernoff's bound and optimization at $\lambda = \rho t/L^2$ now give

$$\begin{aligned} \mathbb{P}(Y \ge t) &\le \inf_{\lambda>0} \exp\left(-\lambda t + \frac{\lambda^2 L^2}{2\rho}\right) \\ &= \exp\left(-\frac{\rho t^2}{2L^2}\right) = \exp\left(-\frac{(r-2)t^2}{96}\right), \end{aligned} \tag{A.91}$$

because $L^2 = 12$ and $\rho = (r-2)/4$. For $r \ge 3$, $r - 2 \ge r/3$, proving the second inequality in equation (A.81). Finally,

$$\exp(-rt^2/288) \le 8\exp(-c_H rt^2) \quad \text{with} \quad c_H = 1/288, \tag{A.92}$$

which proves equation (A.82). The derivation actually holds for all $t > 0$; restricting to $0 < t \le 1$ gives the stated range.

When $d = 0$, $h = 0$ and the left side of equation (A.81) is zero for every $t > 0$, so the conclusion also holds without invoking curvature. No case $X = 0$ is introduced: the row quantifies over nonzero X, which is necessary for the displayed normalization.

□

Proof of the subsection conclusion. Proposition A.10 permits conditioning on the internal shapes while retaining independent Haar $Q_a, Q_b, Q_c \in SO(r)$. For every resulting fixed E and every fixed nonzero X, Lemma A.11 proves

$$\mathbb{E}h = \frac{d}{r^3}$$

by three exact current-notation twirls. Lemma A.12 computes the tensor generator and proves $\mathrm{Lip}(h) \le 2\sqrt{3}$ in the unnormalized product Hilbert–Schmidt geodesic metric. Proposition A.12 computes $\mathrm{Ric} = (r-2)g/4$ in that same metric, applies the metric-matched log-Sobolev inequality and Herbst argument, and obtains

$$\mathbb{P}\left(h \ge \frac{d}{r^3} + t\right) \le \exp(-rt^2/288) \le 8\exp(-c_H rt^2), \qquad c_H = 1/288,$$

for $0 < t \le 1$.

These conclusions are uniform in fixed E, d, X, include $d = 0$ and every actual upper endpoint such as $d = 3kr$, and use only $r \ge 3$ from the large-r setting. The proof introduces no operator norm over targets, net, data-dependent supremum, $O(r)$ surrogate, transformed target, or claim about LQE. Thus Proposition A.12 holds for the fixed objects and with the scope stated there.

□

A.8 Elliptic transfer and raw leverage

Lemma A.13 (Exact oblique-basis projection transfer)**.** *Under Proposition A.11 $\mathscr{S}_0 = LQE$, let $E \subset (\mathbb{R}^r)^{\otimes 3}$ have dimension d, let L be invertible, and, when $d > 0$, let U have orthonormal columns spanning QE. For every coefficient tensor x, define*

$$X = L^\top x, \qquad b = U^\top X.$$

If $d > 0$, then

$$P_{LQE} = LU\big(U^\top L^\top LU\big)^{-1}U^\top L^\top, \tag{A.93}$$

and

$$\|P_{LQE}x\|_F^2 = b^\top\big(U^\top L^\top LU\big)^{-1}b \le \sigma_{\min}(L)^{-2}\|P_{QE}X\|_F^2. \tag{A.94}$$

If $d = 0$, then $QE = LQE = \{0\}$, and both sides of the inequality in equation (A.94) are zero.

Proof. Suppose first that $d > 0$. The columns of LU span LQE. Since L is invertible and U has full column rank, LU has full column rank. The orthogonal projector onto the range of a full-column-rank matrix A is $A(A^\top A)^{-1}A^\top$; substituting $A = LU$ proves equation (A.93) directly.

Because an orthogonal projector is symmetric and idempotent,

$$\begin{aligned} \|P_{LQE}x\|_F^2 &= \langle x, P_{LQE}x\rangle_F \\ &= b^\top\big(U^\top L^\top LU\big)^{-1}b. \end{aligned} \tag{A.95}$$

For every $y \in \mathbb{R}^d$, orthonormality of U gives

$$y^\top U^\top L^\top LUy = \|LUy\|_F^2 \ge \sigma_{\min}(L)^2\|Uy\|_F^2 = \sigma_{\min}(L)^2\|y\|_2^2.$$

Hence

$$\big(U^\top L^\top LU\big)^{-1} \preceq \sigma_{\min}(L)^{-2}I_d. \tag{A.96}$$

Moreover,

$$\|b\|_2^2 = \|U^\top X\|_2^2 = \|UU^\top X\|_F^2 = \|P_{QE}X\|_F^2. \tag{A.97}$$

Combining equations (A.95)–(A.97) proves equation (A.94). When $d = 0$, both subspaces are the zero subspace, so the stated zero-case conclusion is immediate.

□

Proposition A.13 (Elliptic transfer with the exact condition-number loss)**.** *Under the generated event $\mathcal{E}_{\mathrm{cond}}$, the elliptic singular intervals from Lemma A.4, Proposition A.11 $\mathscr{S}_0 = LQE$, and Lemma A.13,*

$$\sigma_{\min}(L) \ge \kappa_1^{-3}, \qquad \sigma_{\max}(L) \le \kappa_1^3. \tag{A.98}$$

Consequently, for every nonzero coefficient tensor x, with $X = L^\top x$,

$$\begin{aligned} \|P_{\mathscr{S}_0}x\|_F^2 &\le \sigma_{\min}(L)^{-2}h(Q;E,X)\|X\|_F^2 \\ &\le \kappa_1^{12}h(Q;E,X)\|x\|_F^2. \end{aligned} \tag{A.99}$$

In particular,

$$\|P_{\mathscr{S}_0}D_r\|_F^2 \le \kappa_1^{12}h(Q;E,L^\top D_r)\|D_r\|_F^2. \tag{A.100}$$

Proof. The singular values of a tensor product are the pairwise products of the singular values of its factors. On $\mathcal{E}_{\text{cond}}$, Lemma A.4 gives $\sigma(H_m) \subset [\kappa_1^{-1}, \kappa_1]$ for $m \in \{a, b, c\}$. Applying these intervals to $L = H_a \otimes H_b \otimes H_c$ gives

$$\sigma_{\min}(L) = \prod_{m \in \{a,b,c\}} \sigma_{\min}(H_m) \geq \kappa_1^{-3},$$

and

$$\sigma_{\max}(L) = \prod_{m \in \{a,b,c\}} \sigma_{\max}(H_m) \leq \kappa_1^{3},$$

proving equation (A.98). These are the simultaneous worst cases in which all three lower singular values equal κ_1^{-1}, or all three upper singular values equal κ_1.

Lemma A.13 and $\|X\|_F \leq \sigma_{\max}(L)\|x\|_F$ give

$$\begin{aligned} \|P_{\mathscr{S}_0} x\|_F^2 &\leq \sigma_{\min}(L)^{-2} \|P_{QE} X\|_F^2 \\ &= \sigma_{\min}(L)^{-2} h(Q; E, X) \|X\|_F^2 \\ &\leq \sigma_{\min}(L)^{-2} \sigma_{\max}(L)^2 h(Q; E, X) \|x\|_F^2 \\ &\leq \kappa_1^{12} h(Q; E, X) \|x\|_F^2, \end{aligned} \tag{A.101}$$

because the lower and upper singular-value factors contribute κ_1^6 each. This proves equation (A.99), and $x = D_r$ gives equation (A.100). If $d = 0$, the same conclusion holds with both projection energies equal to zero.

□

Proposition A.14 (Exact raw target leverage bound)**.** *Under Assumption 3, Proposition A.10, which gives $d \leq 3kr$, Lemma A.11, Proposition A.12, and Proposition A.13, define*

$$\tau_\kappa := \frac{1}{4\kappa_1^{12}}. \tag{A.102}$$

If

$$r \geq \left\lceil (12\kappa_1^{12})^{4/3} \right\rceil, \tag{A.103}$$

then, conditional on every realized factor triple in $\mathcal{E}_{\text{cond}}$,

$$\|P_{\mathscr{S}_0} D_r\|_F^2 \leq \frac{r}{2} \tag{A.104}$$

holds with conditional failure at most

$$8 \exp\left(-\frac{c_H r}{16\kappa_1^{24}} \right). \tag{A.105}$$

Proof. Since $\kappa \geq 1$ and $\kappa_1 = 2\kappa^2 \geq 1$,

$$0 < \tau_\kappa \leq \frac{1}{4} \leq 1, \tag{A.106}$$

so it is an admissible deviation for Proposition A.12. Assumption 3 and the dimension bound give, including the maximal allowed rank,

$$\frac{d}{r^3} \leq \frac{3kr}{r^3} = \frac{3k}{r^2} \leq 3r^{-3/4}. \tag{A.107}$$

The threshold in equation (A.103) is exactly the condition

$$r^{3/4} \geq 12\kappa_1^{12},$$

and therefore

$$3r^{-3/4} \leq \frac{1}{4\kappa_1^{12}} = \tau_\kappa. \tag{A.108}$$

Condition now on the realized factors and the internal shapes. Then E, d are fixed, Q remains product Haar, and $X = L^\mathsf{T} D_r \neq 0$ is fixed. Applying Lemma A.11 and Proposition A.12 with $t = \tau_\kappa$, and using equations (A.107)–(A.108), shows that outside an event of conditional probability at most

$$8e^{-c_H r \tau_\kappa^2} = 8\exp\left(-\frac{c_H r}{16\kappa_1^{24}}\right), \tag{A.109}$$

one has

$$h(Q; E, L^\mathsf{T} D_r) \leq \frac{d}{r^3} + \tau_\kappa \leq 2\tau_\kappa = \frac{1}{2\kappa_1^{12}}. \tag{A.110}$$

Proposition A.13 then gives

$$\begin{aligned} \|P_{\mathscr{S}_0} D_r\|_F^2 &\leq \kappa_1^{12} h(Q; E, L^\mathsf{T} D_r) \|D_r\|_F^2 \\ &\leq \frac{1}{2}\|D_r\|_F^2 = \frac{r}{2}, \end{aligned} \tag{A.111}$$

because the r summands in D_r are orthonormal and hence $\|D_r\|_F^2 = r$. This is exactly (A.104) for the raw target and exact raw tangent span.

The bound in equation (A.109) is uniform in the internal shapes. Integrating over them preserves the same conditional failure probability for every realized factor triple in $\mathcal{E}_{\mathrm{cond}}$; no union bound or operator supremum is used. The case $d = 0$ is deterministic and already satisfies equation (A.111). □

Lemma A.14 (Explicit conversion to the polynomial failure budget)**.** *Let*

$$a_\kappa := \frac{c_H}{16\kappa_1^{24}}, \tag{A.112}$$

and define the explicit threshold

$$r_{0,\mathrm{LEV}}(\kappa) := \left\lceil \max\left\{3, (12\kappa_1^{12})^{4/3}, \left(\frac{20 + \log 8}{a_\kappa}\right)^2\right\}\right\rceil. \tag{A.113}$$

For every integer $r \geq r_{0,\mathrm{LEV}}(\kappa)$,

$$8e^{-a_\kappa r} \leq r^{-20}. \tag{A.114}$$

Proof. For every $r \geq 1$,

$$\log r \leq \sqrt{r}. \tag{A.115}$$

Indeed, $f(r) = \sqrt{r} - \log r$ decreases on $[1, 4]$, increases on $[4, \infty)$, and $f(4) = 2 - \log 4 > 0$. The last term in the maximum in equation (A.113) gives

$$\sqrt{r} \geq \frac{20 + \log 8}{a_\kappa}.$$

Using equation (A.115) and $\sqrt{r} \geq 1$,

$$\begin{aligned} a_\kappa r &= a_\kappa \sqrt{r}\sqrt{r} \\ &\geq (20 + \log 8)\sqrt{r} \\ &\geq 20 \log r + \log 8. \end{aligned} \tag{A.116}$$

Therefore

$$8e^{-a_\kappa r} \leq 8e^{-20\log r - \log 8} = r^{-20},$$

which is equation (A.114). The same threshold includes equation (A.103), so it simultaneously discharges the mean-dominance and tail-conversion conditions.

□

Proof of the subsection conclusion. Lemma A.13 gives the exact orthogonal projection formula for the anisotropic subspace LQE, including the degenerate case $d = 0$, and proves

$$\|P_{LQE}x\|_F^2 \leq \sigma_{\min}(L)^{-2}\|P_{QE}L^\mathsf{T} x\|_F^2.$$

Proposition A.13 applies the singular intervals in all three modes and proves

$$\sigma_{\min}(L) \geq \kappa_1^{-3}, \qquad \sigma_{\max}(L) \leq \kappa_1^3,$$

so the complete worst-case anisotropic loss is exactly κ_1^{12}.

Proposition A.14 cites Assumption 3 to use the maximal-rank inequality

$$\frac{d}{r^3} \leq \frac{3k}{r^2} \leq 3r^{-3/4} \leq \tau_\kappa, \qquad \tau_\kappa = \frac{1}{4\kappa_1^{12}},$$

and applies Lemma A.11 and Proposition A.12 at the single exact tensor $X = L^\mathsf{T} D_r$. It obtains $h \leq 2\tau_\kappa$, hence

$$\|P_{\mathscr{S}_0} D_r\|_F^2 \leq \kappa_1^{12}(2\tau_\kappa)\|D_r\|_F^2 = \frac{r}{2}.$$

Finally, Lemma A.14 gives an explicit $r_{0,\mathrm{LEV}}(\kappa)$ for which the conditional failure in equation (A.105) is at most r^{-20}. Uniformity over the conditioned shapes permits the same bound after integrating them out. These four named results prove that (A.104) holds for the raw D_r and exact raw $\mathscr{S}_0$, with conditional failure at most r^{-20}, without a surrogate residual or target supremum.

□

A.9 Raw tangent-deficit witness

For a realized smoothing triple $(A, B, C) \in \mathcal{E}_{\mathrm{cond}}$, write $\mathbb{P}_{\mathrm{init}}(\cdot \mid A, B, C)$ for conditional probability over the remaining independent raw Gaussian initialization after the matrices A, B, C are fixed. This is precisely the conditional law supplied by Assumption 5; it introduces no independence beyond the setting's initialization–smoothing independence.

Lemma A.15 (Exact membership and nonzero raw normal energy)**.** *Under the setting definitions, Proposition A.8, and Proposition A.14, if* $\|P_{\mathscr{S}_0} D_r\|_F^2 \leq r/2$*, then*

$$\widehat{D}_0 \in \mathscr{S}_0, \qquad \|P_{\mathscr{S}_0^\perp} D_r\|_F^2 \geq \frac{r}{2} > 0. \tag{A.117}$$

Thus the denominator in the setting-required definition of W_0 *is nonzero.*

Proof. The membership in equation (A.117) is exactly the raw membership conclusion of Proposition A.8. The tensors $e_j \otimes e_j \otimes e_j$, $j \in [r]$, are orthonormal, so

$$\|D_r\|_F^2 = \left\| \sum_{j=1}^{r} e_j \otimes e_j \otimes e_j \right\|_F^2 = r. \tag{A.118}$$

Orthogonal Pythagorean decomposition in the raw coefficient space gives

$$\begin{aligned} \|P_{\mathscr{S}_0^\perp} D_r\|_F^2 &= \|D_r\|_F^2 - \|P_{\mathscr{S}_0} D_r\|_F^2 \\ &\geq r - \frac{r}{2} = \frac{r}{2}. \end{aligned} \tag{A.119}$$

Because $r \geq 1$, the right-hand side is strictly positive. Equality in the assumed leverage bound merely makes equation (A.119) an equality and still leaves normal energy $r/2 > 0$. Conversely, a zero denominator would make the left-hand side of equation (A.119) zero, contradicting $r/2 > 0$. □

Proposition A.15 (Raw unit normal witness and tangent deficit)**.** *Under the setting definitions and Lemma A.15, if $\|P_{\mathscr{S}_0} D_r\|_F^2 \leq r/2$, define exactly*

$$W_0 := \frac{P_{\mathscr{S}_0^\perp} D_r}{\|P_{\mathscr{S}_0^\perp} D_r\|_F}. \tag{A.120}$$

Then

$$\|W_0\|_F = 1, \qquad W_0 \perp \mathscr{S}_0, \qquad \langle D_r - \widehat{D}_0, W_0 \rangle \geq \delta_0 \|D_r\|_F, \tag{A.121}$$

where $\delta_0 = 1/8$. Consequently,

$$\left\{ \|P_{\mathscr{S}_0} D_r\|_F^2 \leq \frac{r}{2} \right\} \implies \mathcal{E}_{\mathrm{deficit}}. \tag{A.122}$$

Additionally, under Proposition A.14 and Lemma A.14, suppose $r \geq r_{0,\mathrm{LEV}}(\kappa)$. For every realized $(A, B, C) \in \mathcal{E}_{\mathrm{cond}}$, let $\mathbb{P}_{\mathrm{init}}(\,\cdot \mid A, B, C)$ denote the remaining independent raw-Gaussian initialization law after fixing the realized smoothing matrices. Then

$$\mathbb{P}_{\mathrm{init}}(\mathcal{E}_{\mathrm{deficit}}^{\mathsf{c}} \mid A, B, C) \leq r^{-20}, \tag{A.123}$$

Proof. Lemma A.15 makes the denominator in equation (A.120) strictly positive, so the impossible zero-denominator branch does not occur under the stated leverage bound. By construction, W_0 has unit Frobenius norm and lies in $\mathscr{S}_0^\perp$. The exact membership $\widehat{D}_0 \in \mathscr{S}_0$ gives $\langle \widehat{D}_0, W_0 \rangle = 0$. Therefore

$$\begin{aligned} \langle D_r - \widehat{D}_0, W_0 \rangle &= \left\langle D_r, \frac{P_{\mathscr{S}_0^\perp} D_r}{\|P_{\mathscr{S}_0^\perp} D_r\|_F} \right\rangle_F \\ &= \frac{\|P_{\mathscr{S}_0^\perp} D_r\|_F^2}{\|P_{\mathscr{S}_0^\perp} D_r\|_F} \\ &= \|P_{\mathscr{S}_0^\perp} D_r\|_F \\ &\geq \sqrt{\frac{r}{2}} = \frac{1}{\sqrt{2}} \|D_r\|_F \geq \frac{1}{8} \|D_r\|_F = \delta_0 \|D_r\|_F. \end{aligned} \tag{A.124}$$

The second equality uses the orthogonal decomposition of D_r, equation (A.118) gives $\|D_r\|_F = \sqrt{r}$, and $1/\sqrt{2} \geq 1/8$. Thus equation (A.121) is exactly the witness condition in the setting-defined event, proving equation (A.122).

For every integer $r \geq r_{0,\mathrm{LEV}}(\kappa)$ and every realized factor triple in $\mathcal{E}_{\mathrm{cond}}$, Proposition A.14 supplies the exponential conditional failure bound in equation (A.105), while Lemma A.14 converts that bound to r^{-20} at exactly the same threshold. Combining these two named inputs with the deterministic inclusion in equation (A.122) gives

$$\begin{aligned}\mathbb{P}_{\mathrm{init}}(\mathcal{E}^{c}_{\mathrm{deficit}} \mid A, B, C) &\leq \mathbb{P}_{\mathrm{init}}\left(\|P_{\mathscr{S}_0} D_r\|_F^2 > \frac{r}{2} \,\middle|\, A, B, C\right)\\ &\leq 8\exp\left(-\frac{c_H r}{16\kappa_1^{24}}\right) = 8e^{-a_\kappa r} \\ &\leq r^{-20}.\end{aligned} \tag{A.125}$$

Only the setting's initialization–smoothing independence is used to define this conditional law; no independence between the two displayed events is required, and the deterministic inclusion spends no new failure probability.

□

Proof of the subsection conclusion. Lemma A.15 combines the exact membership $\widehat{D}_0 \in \mathscr{S}_0$, the raw identity $\|D_r\|_F^2 = r$, Pythagoras, and the exact raw leverage bound to prove

$$\|P_{\mathscr{S}_0^\perp} D_r\|_F^2 \geq r/2 > 0.$$

This handles equality in the leverage bound, uses $r \geq 1$, and excludes the only potential zero-denominator branch.

Proposition A.15 then normalizes that exact raw normal component, proves unit norm and orthogonality, cancels $\widehat{D}_0$ by its exact same-span membership, and obtains

$$\langle D_r - \widehat{D}_0, W_0\rangle = \|P_{\mathscr{S}_0^\perp} D_r\|_F \geq \sqrt{r/2} \geq \delta_0 \|D_r\|_F.$$

Hence the raw leverage event is contained in $\mathcal{E}_{\mathrm{deficit}}$ deterministically, and Proposition A.14 together with Lemma A.14 makes equation (A.125) inherit the conditional r^{-20} failure bound without a union bound or additional budget. This proves the claimed raw tangent-deficit bound, with the raw target, raw tangent span, raw represented tensor, raw witness, and Frobenius metric unchanged.

□

A.10 Initialization-event probability

Proposition A.16 (Conditional assembly of the initialization event)**.** *Under Assumptions 1, 2, 3, 4, and 5, and under Propositions A.1, A.4, A.6, and A.15, together with Lemma A.14, if $r \geq r_{0,\mathrm{LEV}}(\kappa)$, then the exact setting event satisfies*

$$\mathbb{P}(\mathcal{E}^{\mathsf{c}}_{\mathrm{init\_norm}}) \leq 4r^{-20}$$

under the joint smoothing/initialization law conditional on the deterministic base triple.

Proof. By the exact event identity,

$$\mathcal{E}_{\mathrm{init\_norm}} = \mathcal{E}_{\mathrm{cond}} \cap \mathcal{E}^{\mathrm{norm}}_{\mathrm{gram}} \cap \mathcal{E}_{\mathrm{size}} \cap \mathcal{E}_{\mathrm{deficit}},$$

and hence

$$\mathcal{E}^{\mathsf{c}}_{\mathrm{init\_norm}} = \mathcal{E}^{\mathsf{c}}_{\mathrm{cond}} \uplus \left[\mathcal{E}_{\mathrm{cond}} \cap \left((\mathcal{E}^{\mathrm{norm}}_{\mathrm{gram}})^{\mathsf{c}} \cup \mathcal{E}^{\mathsf{c}}_{\mathrm{size}} \cup \mathcal{E}^{\mathsf{c}}_{\mathrm{deficit}}\right)\right], \tag{A.126}$$

where $\uplus$ denotes a disjoint union. Let $\mathcal{G} = \sigma(A, B, C)$. Applying the tower property and the conditional union bound to the second term in equation (A.126) gives

$$\begin{aligned}
&\mathbb{P}\big(\mathcal{E}_{\text{cond}} \cap \big((\mathcal{E}_{\text{gram}}^{\text{norm}})^{\mathsf{c}} \cup \mathcal{E}_{\text{size}}^{\mathsf{c}} \cup \mathcal{E}_{\text{deficit}}^{\mathsf{c}}\big)\big) \\
&\quad = \mathbb{E}\big[\mathbf{1}_{\mathcal{E}_{\text{cond}}} \mathbb{P}\big((\mathcal{E}_{\text{gram}}^{\text{norm}})^{\mathsf{c}} \cup \mathcal{E}_{\text{size}}^{\mathsf{c}} \cup \mathcal{E}_{\text{deficit}}^{\mathsf{c}} \big| \mathcal{G}\big)\big] \\
&\quad \leq \mathbb{E}\big[\mathbf{1}_{\mathcal{E}_{\text{cond}}} \left(\mathbb{P}((\mathcal{E}_{\text{gram}}^{\text{norm}})^{\mathsf{c}} \mid \mathcal{G}) + \mathbb{P}(\mathcal{E}_{\text{size}}^{\mathsf{c}} \mid \mathcal{G}) + \mathbb{P}(\mathcal{E}_{\text{deficit}}^{\mathsf{c}} \mid \mathcal{G})\right)\big] \\
&\quad \leq 3r^{-20}.
\end{aligned} \tag{A.127}$$

The last line uses the conditional bounds from Propositions A.4, A.6, and A.15 for every realized triple in $\mathcal{E}_{\text{cond}}$; Lemma A.14 supplies the last proposition's polynomial conversion under the displayed threshold. It does not require independence among the Gram, size, and deficit events or add a fifth failure term. Combining equations (A.126)–(A.127) with the smoothing bound yields

$$\mathbb{P}(\mathcal{E}_{\text{init\_norm}}^{\mathsf{c}}) \leq \mathbb{P}(\mathcal{E}_{\text{cond}}^{\mathsf{c}}) + 3r^{-20} \leq r^{-20} + 3r^{-20} = 4r^{-20}.$$

This proves the proposition.

∎

Proposition A.17 (Public initialization confidence)**.** *Under Assumption 2 and Proposition A.16, if $r \geq 2$, then*

$$\mathbb{P}(\mathcal{E}_{\text{init\_norm}}) \geq 1 - r^{-10}.$$

Proof. For $r \geq 2$,

$$4 \leq 2^{10} \leq r^{10},$$

and therefore

$$4r^{-20} \leq r^{10} r^{-20} = r^{-10}. \tag{A.128}$$

Proposition A.16 and equation (A.128) imply

$$\begin{aligned}
\mathbb{P}(\mathcal{E}_{\text{init_norm}}) &= 1 - \mathbb{P}(\mathcal{E}_{\text{init_norm}}^{\mathsf{c}}) \\
&\geq 1 - 4r^{-20} \\
&\geq 1 - r^{-10}.
\end{aligned}$$

Assumption 2 already permits enlarging the theorem's large-r threshold, so requiring $r \geq 2$ adds no new theorem-facing condition.

∎

Proof of the subsection conclusion. Proposition A.16 uses the exact event identity, the unconditional failure for $\mathcal{E}_{\text{cond}}$, and the three uniformly conditional failure bounds. The tower property and a conditional union bound give

$$\mathbb{P}(\mathcal{E}_{\text{init\_norm}}^{\mathsf{c}}) \leq r^{-20} + 3r^{-20} = 4r^{-20}$$

without multiplying probabilities or asserting independence among the three initialization events. Proposition A.17 then proves explicitly that $4r^{-20} \leq r^{-10}$ for $r \geq 2$, and therefore

$$\boxed{\mathbb{P}(\mathcal{E}_{\text{init\_norm}}) \geq 1 - r^{-10}}.$$

This proves the claim under the exact joint probability space.

∎

A.11 Finite-path convergence and factor radius

Lemma A.16 (finite total variation gives a finite balanced-factor limit)**.** *Under Assumption 6, consider the setting-defined trajectory* $(\theta_t)_{t\ge 0}$*. On the sole local conditional hypothesis* $\mathcal{C}_{\text{path}}$*, define*

$$\ell_t := d_{\text{bal}}(\theta_{t+1}, \theta_t) \ge 0.$$

Then, for every pair of integers $0 \le s < t$*,*

$$d_{\text{bal}}(\theta_s, \theta_t) \le \sum_{u=s}^{t-1} \ell_u. \tag{A.129}$$

The tails $\sum_{u=s}^{\infty} \ell_u$ *tend to zero as* $s \to \infty$*. Consequently, the balanced representatives form a Cauchy sequence in* $(\mathbb{R}^{n\times k})^3$ *and converge in* d_{bal} *to a finite* θ_∞*.*

Proof. Assumption 6 is used only to identify the recursively defined trajectory. The metric formula in the setting is the Euclidean product-space distance, so repeated application of its triangle inequality gives

$$\begin{aligned} d_{\text{bal}}(\theta_s, \theta_t) &\le d_{\text{bal}}(\theta_s, \theta_{s+1}) + d_{\text{bal}}(\theta_{s+1}, \theta_{s+2}) + \cdots + d_{\text{bal}}(\theta_{t-1}, \theta_t) \\ &= \sum_{u=s}^{t-1} \ell_u, \end{aligned}$$

which proves equation (A.129).

Let

$$S_N := \sum_{u=0}^{N-1} \ell_u, \qquad S_0 := 0.$$

On $\mathcal{C}_{\text{path}}$, the nonnegative partial sums increase to the finite value

$$E_{\text{path}} = \sum_{u=0}^{\infty} \ell_u \le E_\star < \infty.$$

Therefore the exact tail remainder

$$R_s := E_{\text{path}} - S_s = \sum_{u=s}^{\infty} \ell_u$$

is nonnegative and satisfies $R_s \downarrow 0$. Given $\varepsilon > 0$, choose N with $R_N < \varepsilon$. For arbitrary $s, t \ge N$, assume without loss of generality that $s < t$. Equation (A.129) and nonnegativity give

$$d_{\text{bal}}(\theta_s, \theta_t) \le \sum_{u=s}^{t-1} \ell_u \le R_s \le R_N < \varepsilon.$$

Thus (θ_t) is Cauchy in the finite-dimensional Euclidean product space $(\mathbb{R}^{n\times k})^3$, which is complete. Hence there is a finite $\theta_\infty = (X_\infty, Y_\infty, Z_\infty)$ such that

$$d_{\text{bal}}(\theta_t, \theta_\infty) \longrightarrow 0.$$

If $E_{\text{path}} = 0$, nonnegativity forces $\ell_t = 0$ for every t. Because d_{bal} is a metric on the displayed product space, $\theta_{t+1} = \theta_t$ for every t, so the trajectory is constant and $\theta_\infty = \theta_0$. At the other boundary $E_{\text{path}} = E_\star$, the same finite-tail argument applies with equality in the total budget and requires no strict slack.

□

Proposition A.18 (horizon-uniform path and endpoint radius)**.**
Under Assumption 6, Lemma A.16, and the sole local conditional hypothesis $\mathcal{C}_{\text{path}}$, every $t \geq 0$ satisfies

$$d_{\text{bal}}(\theta_t, \theta_0) \leq \sum_{u=0}^{t-1} \ell_u \leq E_{\text{path}} \leq E_\star \leq 1, \tag{A.130}$$

where the first sum is empty when $t = 0$. The finite limit produced by Lemma A.16 satisfies

$$d_{\text{bal}}(\theta_\infty, \theta_0) \leq E_{\text{path}} \leq E_\star. \tag{A.131}$$

Proof. Applying equation (A.129) from Lemma A.16 with $s = 0$ gives

$$d_{\text{bal}}(\theta_t, \theta_0) = d_{\text{bal}}(\theta_0, \theta_t) \leq \sum_{u=0}^{t-1} \ell_u \leq \sum_{u=0}^{\infty} \ell_u = E_{\text{path}}.$$

The event $\mathcal{C}_{\text{path}}$ gives $E_{\text{path}} \leq E_\star$. From the setting definition

$$E_\star = \min\left\{1, \sqrt{\frac{\delta_0}{16 C_{\text{CP}}(\kappa, 3)}}\right\},$$

we also have $E_\star \leq 1$. This proves equation (A.130).

Since $\theta_t \to \theta_\infty$ in d_{bal}, continuity of the Euclidean norm, equivalently the reverse triangle inequality, gives

$$d_{\text{bal}}(\theta_\infty, \theta_0) = \lim_{t\to\infty} d_{\text{bal}}(\theta_t, \theta_0).$$

Taking the limit in the already uniform bound $d_{\text{bal}}(\theta_t, \theta_0) \leq E_{\text{path}}$ proves equation (A.131). Both conclusions remain non-strict when $E_{\text{path}} = E_\star$, while the case $E_{\text{path}} = 0$ reduces to the constant trajectory identified in Lemma A.16.

□

Proposition A.19 (uniform component bounds along the path and at the limit)**.** *Under Assumption 6, Proposition A.6, Propositions A.16 and A.17, and Proposition A.18, on $\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}$, for every component $i \in [k]$, every mode $M_t \in \{X_t, Y_t, Z_t\}$, and every $t \geq 0$,*

$$\|m_{i,t}\|_2 \leq 2 + E_\star \leq 3. \tag{A.132}$$

Writing $M_\infty \in \{X_\infty, Y_\infty, Z_\infty\}$ for the corresponding limit block and $m_{i,\infty}$ for its i-th column,

$$\|m_{i,\infty}\|_2 \leq 2 + E_\star \leq 3. \tag{A.133}$$

Proof. By the exact setting identity for $\mathcal{E}_{\text{init\_norm}}$, $\mathcal{E}_{\text{size}}$ holds on $\mathcal{E}_{\text{init\_norm}}$. Hence

$$\|m_{i,0}\|_2 \leq 2$$

for every component and mode.

Fix i, one mode matrix M_t, and $t \geq 0$. The Euclidean norm of one column is bounded by the Frobenius norm of the full mode-matrix difference, and that Frobenius norm is one summand of the product norm defining d_{bal}. Therefore

$$\begin{aligned}
\|m_{i,t}\|_2 &\leq \|m_{i,0}\|_2 + \|m_{i,t} - m_{i,0}\|_2 \\
&\leq \|m_{i,0}\|_2 + \|M_t - M_0\|_F \\
&\leq \|m_{i,0}\|_2 + d_{\text{bal}}(\theta_t, \theta_0) \\
&\leq 2 + E_\star \leq 3,
\end{aligned}$$

where Proposition A.18 supplies the fourth line and the setting definition supplies $E_\star \le 1$. This proves equation (A.132) simultaneously for every finite time.

The product-space convergence from Lemma A.16 implies $M_t \to M_\infty$ in Frobenius norm and hence identifies $m_{i,\infty}$ as the Euclidean limit of the corresponding columns. Using the endpoint bound from Proposition A.18 directly gives

$$\begin{aligned}\|m_{i,\infty}\|_2 &\le \|m_{i,0}\|_2 + \|m_{i,\infty} - m_{i,0}\|_2\\ &\le \|m_{i,0}\|_2 + \|M_\infty - M_0\|_F\\ &\le \|m_{i,0}\|_2 + d_{\mathrm{bal}}(\theta_\infty, \theta_0)\\ &\le 2 + E_\star \le 3,\end{aligned}$$

which proves equation (A.133). At $E_{\mathrm{path}} = 0$, all inequalities reduce to the unchanged initial columns and $\theta_\infty = \theta_0$. At $E_{\mathrm{path}} = E_\star$, the non-strict bound still gives exactly $2 + E_\star \le 3$.

□

Proof of the subsection conclusion. Work on the exact event $\mathcal{E}_{\mathrm{init\_norm}} \cap \mathcal{C}_{\mathrm{path}}$. Assumption 6 identifies the trajectory but supplies no convergence claim. Lemma A.16 uses only the sole local conditional hypothesis $\mathcal{C}_{\mathrm{path}}$ to prove the exact tail-sum bound

$$d_{\mathrm{bal}}(\theta_s, \theta_t) \le \sum_{u=s}^{t-1} \ell_u,$$

shows that the tails vanish, and obtains a finite limit θ_∞ by completeness of the fixed Euclidean product space. Proposition A.18 then proves the horizon-uniform and endpoint bounds

$$d_{\mathrm{bal}}(\theta_t, \theta_0) \le E_\star, \qquad d_{\mathrm{bal}}(\theta_\infty, \theta_0) \le E_\star.$$

The event identity and Proposition A.6 give $\mathcal{E}_{\mathrm{size}}$ on $\mathcal{E}_{\mathrm{init\_norm}}$, and hence the initial column bound 2. Proposition A.19 combines that bound with the exact mode-matrix displacement dominated by d_{bal} to prove, for every component, mode, finite time, and the limit,

$$\|m_{i,t}\|_2 \le 2 + E_\star \le 3, \qquad \|m_{i,\infty}\|_2 \le 2 + E_\star \le 3.$$

Thus the exact conditional path-convergence and radius-3 claim is proved without upgrading $\mathcal{C}_{\mathrm{path}}$ to an unconditional theorem and without inserting convergence, trapping, or loss into the certificate.

□

A.12 Multilinear endpoint remainder

Lemma A.17 (exact trilinear coefficient expansion)**.** *Under Proposition A.1, on $\mathcal{E}_{\mathrm{cond}}$, let $\theta = (X, Y, Z)$ and $\theta' = (X', Y', Z')$, and write*

$$\Delta X = X' - X, \qquad \Delta Y = Y' - Y, \qquad \Delta Z = Z' - Z.$$

For every $i \in [k]$, define the exact raw coefficient columns

$$\alpha_i = A^\dagger x_i, \quad \beta_i = B^\dagger y_i, \quad \gamma_i = C^\dagger z_i$$

and their increments

$$\Delta\alpha_i = A^\dagger(x_i' - x_i), \quad \Delta\beta_i = B^\dagger(y_i' - y_i), \quad \Delta\gamma_i = C^\dagger(z_i' - z_i).$$

Then

$$D\Psi_{A,B,C}(\theta)[\theta' - \theta] = \sum_{i=1}^{k}\big(\Delta\alpha_i \otimes \beta_i \otimes \gamma_i + \alpha_i \otimes \Delta\beta_i \otimes \gamma_i \\ + \alpha_i \otimes \beta_i \otimes \Delta\gamma_i\big), \tag{A.134}$$

and, after subtracting the value and all three derivative terms per component, the remainder is exactly

$$\Psi_{A,B,C}(\theta') - \Psi_{A,B,C}(\theta) - D\Psi_{A,B,C}(\theta)[\theta' - \theta] \\ = \sum_{i=1}^{k}\big(\Delta\alpha_i \otimes \Delta\beta_i \otimes \gamma_i + \Delta\alpha_i \otimes \beta_i \otimes \Delta\gamma_i + \alpha_i \otimes \Delta\beta_i \otimes \Delta\gamma_i + \Delta\alpha_i \otimes \Delta\beta_i \otimes \Delta\gamma_i\big). \tag{A.135}$$

Thus equation (A.135) contains exactly three quadratic tensors and one cubic tensor for each component, and contains no constant or first-order term.

Proof. Linearity of the fixed left inverses gives

$$A^\dagger x_i' = \alpha_i + \Delta\alpha_i, \quad B^\dagger y_i' = \beta_i + \Delta\beta_i, \quad C^\dagger z_i' = \gamma_i + \Delta\gamma_i.$$

Substitution into the exact setting map yields

$$\Psi_{A,B,C}(\theta') = \sum_{i=1}^{k}(\alpha_i + \Delta\alpha_i) \otimes (\beta_i + \Delta\beta_i) \otimes (\gamma_i + \Delta\gamma_i).$$

Expanding each trilinear product gives eight tensors: the base tensor, the three tensors linear in one increment, the three tensors quadratic in two increments, and the tensor cubic in all three increments. The coefficient of a scalar perturbation parameter in the three linear tensors is precisely the Frechet derivative in equation (A.134). Subtracting the base value and equation (A.134) leaves exactly equation (A.135). This is an algebraic identity, with no limiting argument, generic smoothness estimate, or remainder notation.

□

Proposition A.20 (quadratic raw Taylor bound from a base-column radius)**.** *Under Proposition A.1 and Lemma A.17, work on $\mathcal{E}_{\mathrm{cond}}$. Let $R \geq 0$ and suppose only that the columns of the base point $\theta = (X, Y, Z)$ satisfy*

$$\max_{i\in[k]} \max\{\|x_i\|_2, \|y_i\|_2, \|z_i\|_2\} \leq R. \tag{A.136}$$

For any $\theta' = (X', Y', Z')$, put

$$d = d_{\mathrm{bal}}(\theta', \theta) = \big(\|\Delta X\|_F^2 + \|\Delta Y\|_F^2 + \|\Delta Z\|_F^2\big)^{1/2},$$

and assume $0 \leq d \leq 1$. Then the three quadratic families in equation (A.135) each have Frobenius norm at most $\kappa_1^3 R d^2$, the cubic family has Frobenius norm at most $\kappa_1^3 d^3 \leq \kappa_1^3 d^2$, and

$$\big\|\Psi_{A,B,C}(\theta') - \Psi_{A,B,C}(\theta) - D\Psi_{A,B,C}(\theta)[\theta' - \theta]\big\|_F \leq \kappa_1^3(1 + 3R)d^2 = C_{\mathrm{CP}}(\kappa, R)d^2. \tag{A.137}$$

No column-radius condition on θ' is required.

Proof. On $\mathcal{E}_{\mathrm{cond}}$, Proposition A.1 gives

$$\|A^\dagger\|_{\mathrm{op}}, \ \|B^\dagger\|_{\mathrm{op}}, \ \|C^\dagger\|_{\mathrm{op}} \leq \kappa_1. \tag{A.138}$$

Consequently, for every component,

$$\begin{aligned} \|\Delta\alpha_i\|_2 &\le \kappa_1\|\Delta x_i\|_2, \quad \|\alpha_i\|_2 \le \kappa_1 R, \\ \|\Delta\beta_i\|_2 &\le \kappa_1\|\Delta y_i\|_2, \quad \|\beta_i\|_2 \le \kappa_1 R, \\ \|\Delta\gamma_i\|_2 &\le \kappa_1\|\Delta z_i\|_2, \quad \|\gamma_i\|_2 \le \kappa_1 R. \end{aligned} \tag{A.139}$$

Denote the three quadratic sums in their displayed order in equation (A.135) by Q_{xy}, Q_{xz}, Q_{yz}, and denote the cubic sum by C_{xyz}. The triangle inequality, the rank-one Frobenius identity, equation (A.139), and Cauchy–Schwarz give each quadratic family explicitly:

$$\begin{aligned} \|Q_{xy}\|_F &\le \sum_{i=1}^{k} \|\Delta\alpha_i\|_2\|\Delta\beta_i\|_2\|\gamma_i\|_2 \\ &\le \kappa_1^3 R \sum_{i=1}^{k} \|\Delta x_i\|_2\|\Delta y_i\|_2 \le \kappa_1^3 R\|\Delta X\|_F\|\Delta Y\|_F \le \kappa_1^3 R d^2, \end{aligned} \tag{A.140}$$

$$\begin{aligned} \|Q_{xz}\|_F &\le \sum_{i=1}^{k} \|\Delta\alpha_i\|_2\|\beta_i\|_2\|\Delta\gamma_i\|_2 \\ &\le \kappa_1^3 R \sum_{i=1}^{k} \|\Delta x_i\|_2\|\Delta z_i\|_2 \le \kappa_1^3 R\|\Delta X\|_F\|\Delta Z\|_F \le \kappa_1^3 R d^2, \end{aligned} \tag{A.141}$$

$$\begin{aligned} \|Q_{yz}\|_F &\le \sum_{i=1}^{k} \|\alpha_i\|_2\|\Delta\beta_i\|_2\|\Delta\gamma_i\|_2 \\ &\le \kappa_1^3 R \sum_{i=1}^{k} \|\Delta y_i\|_2\|\Delta z_i\|_2 \le \kappa_1^3 R\|\Delta Y\|_F\|\Delta Z\|_F \le \kappa_1^3 R d^2. \end{aligned} \tag{A.142}$$

For the cubic family, let $a_i = \|\Delta x_i\|_2$, $b_i = \|\Delta y_i\|_2$, and $c_i = \|\Delta z_i\|_2$. Two finite-sum inequalities give

$$\begin{aligned} \sum_{i=1}^{k} a_i b_i c_i &\le \left(\sum_i a_i^2\right)^{1/2} \left(\sum_i b_i^2 c_i^2\right)^{1/2} \\ &\le \left(\sum_i a_i^2\right)^{1/2} \left(\sum_i b_i^2\right)^{1/2} \left(\sum_i c_i^2\right)^{1/2}, \end{aligned} \tag{A.143}$$

where the second line also follows directly from $\sum_i b_i^2 c_i^2 \le (\sum_i b_i^2)(\sum_i c_i^2)$. Therefore

$$\begin{aligned} \|C_{xyz}\|_F &\le \kappa_1^3 \sum_{i=1}^{k} a_i b_i c_i \\ &\le \kappa_1^3\|\Delta X\|_F\|\Delta Y\|_F\|\Delta Z\|_F \le \kappa_1^3 d^3 \le \kappa_1^3 d^2, \end{aligned} \tag{A.144}$$

where the last inequality uses exactly $d \le 1$. Applying the triangle inequality to the four exact families in equation (A.135) and using equations (A.140)–(A.144) yields

$$3\kappa_1^3 R d^2 + \kappa_1^3 d^2 = \kappa_1^3(1 + 3R)d^2,$$

which is equation (A.137).

Only the unperturbed coefficients $\alpha_i, \beta_i, \gamma_i$ occur in the three quadratic terms, one per term. The comparison point θ' enters exclusively through $\Delta X, \Delta Y, \Delta Z$. This is why equation (A.136) is needed only at the base point and no radius for θ' is used.

If $d = 0$, all three physical increments and hence all coefficient increments vanish, so equation (A.135) is exactly zero and equation (A.137) holds with equality. If $d = 1$, the only degree reduction used above is $d^3 \le d^2$, which is equality at that boundary; all other estimates remain non-strict. Thus both boundary values are included.

$\square$

Proposition A.21 (one-endpoint raw coefficient remainder)**.** *Under Proposition A.1, Lemma A.16 and Propositions A.18 and A.19, and Proposition A.20, on the exact conditional event $\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}$,*

$$\begin{aligned} &\|\Psi_{A,B,C}(\theta_\infty) - \Psi_{A,B,C}(\theta_0) - D\Psi_{A,B,C}(\theta_0)[\theta_\infty - \theta_0]\|_F \\ &\quad \le C_{\text{CP}}(\kappa, 3) d_{\text{bal}}(\theta_\infty, \theta_0)^2 \le C_{\text{CP}}(\kappa, 3) E_\star^2 \le \frac{\delta_0}{16}. \end{aligned} \tag{A.145}$$

This is a raw coefficient-space Frobenius bound for the single endpoint pair; it makes no physical-coordinate change and contains no sum over time.

Proof. On $\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}$, Proposition A.19 gives

$$\max_{i\in[k]} \max\{\|x_{i,0}\|_2, \|y_{i,0}\|_2, \|z_{i,0}\|_2\} \le 3. \tag{A.146}$$

This is the required base-point condition with $R = 3$. No endpoint column bound is needed for Proposition A.20. Proposition A.18 gives

$$d_{\text{bal}}(\theta_\infty, \theta_0) \le E_{\text{path}} \le E_\star \le 1. \tag{A.147}$$

Moreover, the definition of $\mathcal{E}_{\text{init\_norm}}$ includes $\mathcal{E}_{\text{cond}}$, so the left-inverse bounds hold for the same realization. Proposition A.20, with $\theta = \theta_0$, $\theta' = \theta_\infty$, $R = 3$, and equation (A.147), gives the first two inequalities in equation (A.145). Finally, the exact setting definition

$$E_\star = \min\left\{1, \sqrt{\frac{\delta_0}{16 C_{\text{CP}}(\kappa, 3)}}\right\}$$

implies

$$C_{\text{CP}}(\kappa, 3) E_\star^2 \le \frac{\delta_0}{16},$$

which proves the last inequality.

At the zero-path boundary, Lemma A.16 gives $\theta_\infty = \theta_0$, and every term on the left of equation (A.145) vanishes exactly. At the unit-displacement boundary, the generic estimate already includes $d = 1$; no strict radius or displacement inequality is used.

$\square$

Proof of the subsection conclusion. Lemma A.17 expands the exact setting map $\Psi_{A,B,C}$ in the raw coefficient coordinates induced by the realized left inverses. After removing $\Psi_{A,B,C}(\theta)$ and the exact Frechet derivative, it leaves exactly three quadratic tensors and one cubic tensor per component. Proposition A.20 applies the κ_1 left-inverse bounds, the base physical column radius, the rank-one Frobenius identity, and finite-sum Cauchy–Schwarz. Each quadratic family is bounded by $\kappa_1^3 R d^2$, while the cubic family is bounded by $\kappa_1^3 d^3 \le \kappa_1^3 d^2$. Their sum is exactly

$$C_{\text{CP}}(\kappa, R) d^2 = \kappa_1^3 (1 + 3R) d^2.$$

On $\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}$, Lemma A.16 supplies the finite endpoint, Proposition A.18 supplies $d_{\text{bal}}(\theta_\infty, \theta_0) \le E_\star \le 1$, and Proposition A.19 supplies the base radius $R = 3$. Since this event is contained in $\mathcal{E}_{\text{cond}}$, Proposition A.1 supplies the required left-inverse bounds. Proposition A.21 therefore establishes the exact one-endpoint raw coefficient remainder

$$\|\Psi_{A,B,C}(\theta_\infty) - \Psi_{A,B,C}(\theta_0) - D\Psi_{A,B,C}(\theta_0)[\theta_\infty - \theta_0]\|_F \le C_{\mathrm{CP}}(\kappa, 3)E_\star^2 \le \delta_0/16.$$

This is the claimed Taylor constant. It is a single endpoint estimate, not a time accumulation, and it remains in the exact raw coefficient Frobenius norm.

□

A.13 Persistence of the raw coefficient margin

Lemma A.18 (raw tangent-range cancellation at initialization)**.** *Under the exact setting definitions and Lemma A.16, Proposition A.18, and A.21, work on $\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}$. Since this event is contained in $\mathcal{E}_{\text{deficit}}$, select W_0 directly from the existential clause defining that event, so that*

$$\|W_0\|_F = 1, \qquad W_0 \perp \mathscr{S}_0, \qquad \langle D_r - \widehat{D}_0, W_0\rangle_F \ge \delta_0\|D_r\|_F = \delta_0\sqrt{r}. \tag{A.148}$$

Let

$$\Delta\theta = \theta_\infty - \theta_0 = (\Delta X, \Delta Y, \Delta Z).$$

Then

$$D\Psi_{A,B,C}(\theta_0)[\Delta\theta] \in \mathscr{S}_0, \qquad \langle D\Psi_{A,B,C}(\theta_0)[\Delta\theta], W_0\rangle_F = 0. \tag{A.149}$$

Proof. Write $\Delta x_i, \Delta y_i, \Delta z_i$ for the columns of the three increment matrices. The exact derivative formula from Lemma A.17, specialized at θ_0, is

$$\begin{aligned} D\Psi_{A,B,C}(\theta_0)[\Delta\theta] = \sum_{i=1}^{k} \big(&(A^\dagger \Delta x_i) \otimes \beta_{i,0} \otimes \gamma_{i,0} \\ &+ \alpha_{i,0} \otimes (B^\dagger \Delta y_i) \otimes \gamma_{i,0} \\ &+ \alpha_{i,0} \otimes \beta_{i,0} \otimes (C^\dagger \Delta z_i) \big). \end{aligned} \tag{A.150}$$

Each of $A^\dagger \Delta x_i$, $B^\dagger \Delta y_i$, and $C^\dagger \Delta z_i$ lies in $\mathbb{R}^r$. By the exact raw tangent definition in Section 1, the three summands in equation (A.150) belong, respectively, to the three generator families whose span is $\mathscr{S}_0$. Their finite sum therefore lies in $\mathscr{S}_0$. The event-wide selection in (A.148) gives $W_0 \perp \mathscr{S}_0$, which proves the second equality in equation (A.149). No quotient representative, normalized target, or closure of the tangent space is used.

□

Proposition A.22 (preserved positive raw coefficient margin)**.** *Under the exact setting definitions, Lemma A.16, Proposition A.18, and A.21, and Lemma A.18, on $\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}$, use the same event-wide witness W_0 selected in (A.148) and define the exact endpoint remainder*

$$R_\infty := \Psi_{A,B,C}(\theta_\infty) - \Psi_{A,B,C}(\theta_0) - D\Psi_{A,B,C}(\theta_0)[\Delta\theta]. \tag{A.151}$$

Then

$$\langle D_r - \Psi_{A,B,C}(\theta_\infty), W_0\rangle_F \ge \frac{15}{16}\delta_0\sqrt{r}, \tag{A.152}$$

and consequently

$$\|D_r - \Psi_{A,B,C}(\theta_\infty)\|_F \ge \frac{15}{16}\delta_0\sqrt{r}. \tag{A.153}$$

Proof. Rearranging equation (A.151) gives the exact raw coefficient identity

$$D_r - \Psi_{A,B,C}(\theta_\infty) = D_r - \Psi_{A,B,C}(\theta_0) - D\Psi_{A,B,C}(\theta_0)[\Delta\theta] - R_\infty. \tag{A.154}$$

By the setting definitions,

$$\Psi_{A,B,C}(\theta_0) = \sum_{i=1}^{k} \alpha_{i,0} \otimes \beta_{i,0} \otimes \gamma_{i,0} = \widehat{D}_0. \tag{A.155}$$

Pairing equation (A.154) with W_0, using the exact cancellation in Lemma A.18, the initial margin and unit norm in (A.148), and Frobenius Cauchy–Schwarz, gives

$$\begin{aligned} \langle D_r - \Psi_{A,B,C}(\theta_\infty), W_0\rangle_F &= \langle D_r - \widehat{D}_0, W_0\rangle_F - \langle R_\infty, W_0\rangle_F \\ &\geq \delta_0\sqrt{r} - |\langle R_\infty, W_0\rangle_F| \\ &\geq \delta_0\sqrt{r} - \|R_\infty\|_F. \end{aligned} \tag{A.156}$$

No sign is assigned to the remainder pairing. The endpoint estimate in Proposition A.21, specifically (A.145), and $r \geq 1$ imply

$$\|R_\infty\|_F \leq \frac{\delta_0}{16} \leq \frac{\delta_0\sqrt{r}}{16}. \tag{A.157}$$

Substituting equation (A.157) into equation (A.156) proves equation (A.152). Finally, Frobenius Cauchy–Schwarz and $\|W_0\|_F = 1$ give

$$\|D_r - \Psi_{A,B,C}(\theta_\infty)\|_F \geq \left|\langle D_r - \Psi_{A,B,C}(\theta_\infty), W_0\rangle_F\right| \geq \frac{15}{16}\delta_0\sqrt{r},$$

because the pairing is positive by equation (A.152). This proves equation (A.153).

If $E_{\text{path}} = 0$, Lemma A.16 gives $\theta_\infty = \theta_0$, so $\Delta\theta = 0$ and $R_\infty = 0$. Equation (A.154) then retains the complete initial margin $\langle D_r - \widehat{D}_0, W_0\rangle_F \geq \delta_0\sqrt{r}$, rather than only the conservative factor $15/16$. If $E_{\text{path}} = E_\star$, the non-strict bound in (A.145) still gives equation (A.157). If the initial deficit inequality in (A.148) holds with equality, equation (A.156) still yields the claimed floor after the worst-sign remainder. For $r = 1$, the second inequality in equation (A.157) is equality, so the same $15/16$ conclusion holds without a large-r simplification.

□

Proof of the subsection conclusion. Lemma A.18 expands the exact raw coefficient derivative at θ_0 and places each of its three families inside a defining generator family of $\mathscr{S}_0$. The raw witness is orthogonal to this exact span, so the complete first-order term cancels.

Proposition A.22 then uses the exact identity $\Psi_{A,B,C}(\theta_0) = \widehat{D}_0$, the initial raw margin, and the single-endpoint remainder. It controls the remainder pairing by absolute value and proves

$$\langle D_r - \Psi_{A,B,C}(\theta_\infty), W_0\rangle_F \geq \delta_0\sqrt{r} - \frac{\delta_0}{16} \geq \frac{15}{16}\delta_0\sqrt{r}.$$

Unit norm of W_0 converts this positive pairing into the same lower bound for the exact raw coefficient residual norm. The argument is deterministic on $\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}$, uses one endpoint remainder rather than a time sum, and makes no physical-coordinate transfer.

□

A.14 Same-target transfer to physical loss

Lemma A.19 (exact mode-span projection of the physical residual)**.** *Under the basic setting definitions and Proposition A.1, work on $\mathcal{E}_{\text{cond}}$. For every $\theta = (X, Y, Z)$, let*

$$\mathcal{P} := P_A \otimes P_B \otimes P_C, \qquad \mathcal{L} := A \otimes B \otimes C.$$

Then $\mathcal{P}$ is the orthogonal projection onto $\operatorname{range}(A) \otimes \operatorname{range}(B) \otimes \operatorname{range}(C)$*,*

$$\mathcal{P}(T - S(\theta)) = \mathcal{L}(D_r - \Psi_{A,B,C}(\theta)), \tag{A.158}$$

and

$$\|T - S(\theta)\|_F^2 = \|\mathcal{P}(T - S(\theta))\|_F^2 + \|(I - \mathcal{P})(T - S(\theta))\|_F^2. \tag{A.159}$$

Proof. Proposition A.1 gives full column rank for every mode matrix on $\mathcal{E}_{\text{cond}}$, so

$$P_M = M(M^\mathsf{T} M)^{-1} M^\mathsf{T}$$

is self-adjoint and idempotent, satisfies $P_M M = M$, and obeys $P_M x = M(M^\dagger x)$. Tensoring self-adjointness and idempotence shows that $\mathcal{P}$ is the stated orthogonal projection.

Because the target factors are columns of A, B, C,

$$\mathcal{P}T = T = \mathcal{L}D_r. \tag{A.160}$$

For the model columns, with $\alpha_i = A^\dagger x_i$, $\beta_i = B^\dagger y_i$, and $\gamma_i = C^\dagger z_i$,

$$\begin{aligned} \mathcal{P}S(\theta) &= \sum_{i=1}^{k} P_A x_i \otimes P_B y_i \otimes P_C z_i \\ &= \sum_{i=1}^{k} A\alpha_i \otimes B\beta_i \otimes C\gamma_i = \mathcal{L}\Psi_{A,B,C}(\theta). \end{aligned} \tag{A.161}$$

Subtracting (A.161) from (A.160) proves (A.158). The tensor in parentheses on the right of (A.158) lies in $(\mathbb{R}^r)^{\otimes 3}$, while both sides lie in $(\mathbb{R}^n)^{\otimes 3}$, so the mode dimensions agree.

Finally, $\mathcal{P}(T - S(\theta))$ and $(I - \mathcal{P})(T - S(\theta))$ are orthogonal because $\mathcal{P}(I - \mathcal{P}) = 0$. Pythagoras gives (A.159). Thus the only discarded quantity is the explicitly nonnegative second squared norm in (A.159).

□

Proposition A.23 (positive relative physical loss)**.**
Under Propositions A.1 and A.22, and Lemma A.19, work on $\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}$. Then, for every $H \in (\mathbb{R}^r)^{\otimes 3}$,

$$\kappa_1^{-3}\|H\|_F \le \|(A \otimes B \otimes C)H\|_F \le \kappa_1^3\|H\|_F. \tag{A.162}$$

Moreover,

$$0 < \kappa_1^{-3}\sqrt{r} \le \|T\|_F \le \kappa_1^3\sqrt{r}, \tag{A.163}$$

$$\|T - S(\theta_\infty)\|_F \ge \frac{15}{16}\delta_0\kappa_1^{-6}\|T\|_F, \tag{A.164}$$

and, with $\epsilon_0(\kappa) = ((15/16)\delta_0)^2\kappa_1^{-12} > 0$,

$$F(\theta_\infty) \ge \epsilon_0(\kappa)\|T\|_F^2 > 0. \tag{A.165}$$

Proof. Take compact singular-value decompositions $M = U_M \Sigma_M V_M^\mathsf{T}$ for $M = A, B, C$. Then

$$\begin{aligned} A \otimes B \otimes C = (U_A \otimes U_B \otimes U_C)(\Sigma_A \otimes \Sigma_B \otimes \Sigma_C) \\ \cdot (V_A \otimes V_B \otimes V_C)^\mathsf{T}. \end{aligned} \tag{A.166}$$

The first factor is an isometry into the physical tensor space, the last is orthogonal, and the middle diagonal entries are $\sigma_i(A)\sigma_j(B)\sigma_\ell(C)$. Proposition A.1 places every one-mode singular value in $[\kappa_1^{-1}, \kappa_1]$; multiplying the three factors gives (A.162), including the exact lower product κ_1^{-3} and upper product κ_1^3.

The r tensors $e_j \otimes e_j \otimes e_j$ are orthonormal, so $\|D_r\|_F = \sqrt{r}$. Since $T = (A \otimes B \otimes C)D_r$, equation (A.162) gives

$$0 < \kappa_1^{-3}\sqrt{r} \le \|T\|_F \le \kappa_1^3 \sqrt{r}. \tag{A.167}$$

In particular, the target upper bound holds and the target is nonzero, which proves (A.163).

Apply (A.158)–(A.159) at $\theta = \theta_\infty$, then (A.162), and finally the raw margin from Proposition A.22, namely (A.153):

$$\begin{aligned} \|T - S(\theta_\infty)\|_F &\ge \|\mathcal{P}(T - S(\theta_\infty))\|_F \\ &= \|(A \otimes B \otimes C)(D_r - \Psi_{A,B,C}(\theta_\infty))\|_F \\ &\ge \kappa_1^{-3}\|D_r - \Psi_{A,B,C}(\theta_\infty)\|_F \\ &\ge \kappa_1^{-3}\frac{15}{16}\delta_0\sqrt{r}. \end{aligned} \tag{A.168}$$

Combining (A.167) and (A.168) in the required direction gives

$$\|T - S(\theta_\infty)\|_F \ge \frac{15}{16}\delta_0 \kappa_1^{-6}\|T\|_F. \tag{A.169}$$

This proves (A.164). Both sides are nonnegative. Squaring (A.169) and using the exact definition of F proves (A.165). Since $\delta_0 = 1/8$ and $\kappa_1 = 2\kappa^2 > 0$, $\epsilon_0(\kappa) > 0$; (A.167) then makes the right side of (A.165) strictly positive.

At equality in the allowed singular bounds, all displayed inequalities remain valid with a positive lower product. If the orthogonal term in (A.159) vanishes, the projected term still has the floor (A.168); if it is nonzero, it only increases the physical norm. A zero raw residual is excluded by Proposition A.22, and a zero target is excluded by (A.167). Zero model columns cause no exception because (A.160)–(A.161) use only linear projections. At $E_{\text{path}} = 0$, Proposition A.22 retains the full raw margin $\delta_0\sqrt{r}$, so the same calculation preserves the stronger baseline constant $\delta_0^2\kappa_1^{-12}$. □

Proof of the subsection conclusion. Lemma A.19 proves the exact projected residual identity directly for the actual residual and shows by (A.159) that projection discards only a nonnegative orthogonal squared norm. Proposition A.23 combines the raw floor with the exact tensor singular products, proves the target and relative-residual bounds, checks the normalization direction before squaring, and proves the positive physical-loss floor. The event scope, target, residual, Frobenius norm, and conditional mode are unchanged. □

A.15 Continuity and conditional probability

Proposition A.24 (polynomial continuity of the CP model and objective)**.** *Under the basic setting definitions, fix any realization of the target T. The map*

$$S : (\mathbb{R}^{n\times k})^3 \longrightarrow (\mathbb{R}^n)^{\otimes 3}$$

is a homogeneous polynomial map of degree three, and $F : (\mathbb{R}^{n\times k})^3 \to \mathbb{R}$ is a polynomial of degree at most six. Both maps are continuous in the Euclidean product metric d_{bal}. In particular, if $d_{\mathrm{bal}}(\theta_t, \theta_\infty) \to 0$, then

$$\|S(\theta_t) - S(\theta_\infty)\|_F \to 0, \qquad F(\theta_t) \to F(\theta_\infty). \tag{A.170}$$

Proof. Write tensor coordinates as $1 \le a, b, c \le n$. From the setting definition,

$$[S(X, Y, Z)]_{abc} = \sum_{i=1}^{k} X_{ai} Y_{bi} Z_{ci}. \tag{A.171}$$

Thus every coordinate of S is a finite sum of degree-three monomials in the entries of (X, Y, Z). Hence S is a homogeneous polynomial map of degree three.

For the fixed realized target, the objective has the coordinate expression

$$F(X, Y, Z) = \sum_{a=1}^{n} \sum_{b=1}^{n} \sum_{c=1}^{n} \left(T_{abc} - \sum_{i=1}^{k} X_{ai} Y_{bi} Z_{ci} \right)^2 . \tag{A.172}$$

Each summand is the square of a polynomial of degree at most three, so (A.172) is a polynomial of degree at most six. Coordinate projections are continuous, and finite sums and products of continuous real-valued functions are continuous. Equations (A.171)–(A.172), together with the finite number of tensor coordinates, therefore prove continuity of both maps.

Finally, the setting identity

$$d_{\mathrm{bal}}(\theta, \theta') = \left(\|X - X'\|_F^2 + \|Y - Y'\|_F^2 + \|Z - Z'\|_F^2 \right)^{1/2}$$

is exactly the Euclidean norm distance on the parameter space. Applying the just-proved continuity to $d_{\mathrm{bal}}(\theta_t, \theta_\infty) \to 0$ proves (A.170).

□

Proposition A.25 (endpoint continuity and exact event inclusion)**.**
Under Assumption 6, Lemma A.16, Proposition A.23, and Proposition A.24, every outcome in $\mathcal{E}_{\mathrm{init\_norm}} \cap \mathcal{C}_{\mathrm{path}}$ satisfies

$$d_{\mathrm{bal}}(\theta_t, \theta_\infty) \to 0, \qquad \lim_{t\to\infty} F(\theta_t) = F(\theta_\infty) \ge \epsilon_0(\kappa)\|T\|_F^2 > 0. \tag{A.173}$$

Consequently,

$$\mathcal{E}_{\mathrm{init\_norm}} \cap \mathcal{C}_{\mathrm{path}} \subseteq \mathcal{F}_+. \tag{A.174}$$

Proof. Fix an arbitrary outcome in $\mathcal{E}_{\mathrm{init\_norm}} \cap \mathcal{C}_{\mathrm{path}}$. Lemma A.16 produces a finite θ_∞ and proves $d_{\mathrm{bal}}(\theta_t, \theta_\infty) \to 0$. Applying Proposition A.24 to this exact convergence yields

$$\lim_{t\to\infty} F(\theta_t) = F(\theta_\infty). \tag{A.175}$$

On the same outcome, Proposition A.23 gives

$$F(\theta_\infty) \ge \epsilon_0(\kappa)\|T\|_F^2 > 0. \tag{A.176}$$

Combining (A.175)–(A.176) proves (A.173).

The setting-defined event $\mathcal{F}_+$ is precisely the event that the balanced-GD representatives converge in d_{bal} to a finite limit and that their objective limit obeys the positive relative floor in (A.173). Thus (A.173) is exactly membership in $\mathcal{F}_+$, not membership in a surrogate endpoint or finite-time event. Since the chosen outcome was arbitrary, (A.174) follows.

□

Proposition A.26 (exact conditional-probability accounting)**.** *Under Assumption 2, Proposition A.17, and Proposition A.25, work under the setting's joint smoothing/initialization law conditional on an arbitrary admissible deterministic base triple. For $r \geq 2$, $\mathbb{P}(\mathcal{E}_{\text{init\_norm}}) > 0$, the conditional probability $\mathbb{P}(\mathcal{C}_{\text{path}} \mid \mathcal{E}_{\text{init\_norm}})$ is well defined by event conditioning, and*

$$\begin{aligned} \mathbb{P}(\mathcal{F}_+) &\geq \mathbb{P}(\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}), \\ \mathbb{P}(\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}) &= \mathbb{P}(\mathcal{E}_{\text{init\_norm}})\mathbb{P}(\mathcal{C}_{\text{path}} \mid \mathcal{E}_{\text{init\_norm}}), \\ \mathbb{P}(\mathcal{F}_+) &\geq (1 - r^{-10})\mathbb{P}(\mathcal{C}_{\text{path}} \mid \mathcal{E}_{\text{init\_norm}}). \end{aligned}$$

All three relations remain valid when the final conditional probability is zero.

Proof. Proposition A.17 gives

$$\mathbb{P}(\mathcal{E}_{\text{init\_norm}}) \geq 1 - r^{-10}.$$

For $r \geq 2$, $r^{-10} < 1$, and therefore

$$\mathbb{P}(\mathcal{E}_{\text{init\_norm}}) \geq 1 - r^{-10} > 0. \tag{A.177}$$

Assumption 2 permits enlarging the sufficiently-large r threshold to include $r \geq 2$, so (A.177) adds no theorem-facing restriction. It also verifies the positive denominator before conditioning.

By Proposition A.25 and monotonicity of probability,

$$\mathbb{P}(\mathcal{F}_+) \geq \mathbb{P}(\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}). \tag{A.178}$$

Using (A.177), the definition of conditional probability for the two exact setting events gives the identity

$$\mathbb{P}(\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}) = \mathbb{P}(\mathcal{E}_{\text{init\_norm}})\mathbb{P}(\mathcal{C}_{\text{path}} \mid \mathcal{E}_{\text{init\_norm}}). \tag{A.179}$$

The conditional probability in (A.179) is nonnegative. Multiplying the lower bound for $\mathbb{P}(\mathcal{E}_{\text{init\_norm}})$ by this nonnegative factor preserves the inequality direction:

$$\mathbb{P}(\mathcal{E}_{\text{init\_norm}})\mathbb{P}(\mathcal{C}_{\text{path}} \mid \mathcal{E}_{\text{init\_norm}}) \geq (1 - r^{-10})\mathbb{P}(\mathcal{C}_{\text{path}} \mid \mathcal{E}_{\text{init\_norm}}). \tag{A.180}$$

Equations (A.178)–(A.180) prove the three relations displayed in the proposition.

If $\mathbb{P}(\mathcal{C}_{\text{path}} \mid \mathcal{E}_{\text{init\_norm}}) = 0$, equation (A.179) says that the intersection has probability zero and (A.180) has zero on both sides. Thus the theorem still asserts only $\mathbb{P}(\mathcal{F}_+) \geq 0$; it does not acquire an unconditional positive failure probability or any lower bound on the path event. No independence assertion is used anywhere.

□

Proof of the subsection conclusion. Proposition A.24 proves directly from the CP coordinates that S and F are finite-dimensional polynomial maps and converts the d_{bal}-limit into

$$\lim_{t \to \infty} F(\theta_t) = F(\theta_\infty).$$

Lemma A.16 supplies that finite limit on the exact path event, while Proposition A.23 supplies the actual physical endpoint floor. Proposition A.25 combines them to obtain the exact inclusion

$$\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}} \subseteq \mathcal{F}_+.$$

Proposition A.17 and the explicit large-r check in Proposition A.26 show that the conditioning event has positive probability. The same proposition then uses only event monotonicity and the definition of conditional probability to prove

$$\boxed{\mathbb{P}(\mathcal{F}_+) \geq (1 - r^{-10})\mathbb{P}(\mathcal{C}_{\text{path}} \mid \mathcal{E}_{\text{init\_norm}})}.$$

The probability space, all-time/asymptotic mode, physical objective, and setting events are unchanged, and the unresolved conditional factor is retained exactly even at its zero boundary.

□

A.16 Proof of the Main Theorem

Proof of Theorem 3.1. Proposition A.17 proves $\mathbb{P}(\mathcal{E}_{\text{init\_norm}}) \geq 1 - r^{-10}$. Lemma A.16 and Propositions A.23 and A.25 prove, on $\mathcal{E}_{\text{init\_norm}} \cap \mathcal{C}_{\text{path}}$, the finite parameter limit and the positive asymptotic physical-loss floor with $\epsilon_0(\kappa) = ((15/16)\delta_0)^2\kappa_1^{-12}$. Finally, Proposition A.26 proves the exact event-conditioning identity and

$$\mathbb{P}(\mathcal{F}_+) \geq (1 - r^{-10})\mathbb{P}(\mathcal{C}_{\text{path}} \mid \mathcal{E}_{\text{init\_norm}}).$$

That proposition also verifies the positive conditioning denominator and retains the conditional factor when it equals zero. These are precisely all claims in Theorem 3.1. ☐

References

Bakry, D., Gentil, I. and Ledoux, M. (2014). *Analysis and Geometry of Markov Diffusion Operators*. Springer, Cham. 30

Besse, A. L. (1987). *Einstein Manifolds*. Springer-Verlag, Berlin. 30

Davidson, K. R. and Szarek, S. J. (2001). Local operator theory, random matrices and banach spaces. In *Handbook of the Geometry of Banach Spaces* (W. B. Johnson and J. Lindenstrauss, eds.), vol. 1. North-Holland, Amsterdam, 317–366. 6, 13

Folland, G. B. (1999). *A Course in Abstract Harmonic Analysis*. CRC Press, Boca Raton, FL. 24

Laurent, B. and Massart, P. (2000). Adaptive estimation of a quadratic functional by model selection. *The Annals of Statistics* **28** 1302–1338. 12, 13

Stewart, G. W. and Sun, J.-g. (1990). *Matrix Perturbation Theory*. Academic Press, Boston. 7

Tropp, J. A. (2012). User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics* **12** 389–434. 14